

DOT E 6635: Artificial Intelligence for Business Research (Spring 2026)

Deep Reinforcement Learning

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Where Are We?

- Model-based methods
 - We know the action space, the state space, the transition probability matrix, and the reward function.
 - Bellman equation to evaluate a policy and Bellman optimality operator to find the optimal policy.
 - Value iteration, policy iteration, linear programming.
- Model-free methods
 - Monte Carlo (MC) learns directly from episodes of experience: Sample the entire reward process and use empirical mean return to approximate expected return.
 - Temporal Difference (TD) combines MC and Bellman operator: Bootstrap to update the value function based on the existing estimate.
 - MC has zero bias and high variance; TD has some bias and low variance.
- Model-free control
 - MC policy iteration combines MC policy evaluation and ϵ -greedy for policy improvement.
 - SARSA is an on-policy method to update $Q(S,A)$ using TD and ϵ -greedy for policy improvement.
 - Q-learning is an off-policy method to learn $Q(S,A)$ for the greedy target policy using ϵ -greedy as the behavior policy through TD update.

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Agenda

- Value Function Approximation
- DQN
- Policy-based Methods
- Modern DRL
- RL for LLM

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Value Function Approximation

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-6-value-function-approximation-.pdf>

- So far we assume that we can represent the MDP using **state and action pairs** (Q-function).
 - This is called **tabular RL**.
- What if the state or action spaces are **too large**?
 - Continuous states: self-driving car
 - Large states: Go has $3^{19 \times 19} \approx 10^{137}$ states (# of atoms in universe: $\sim 10^{80}$)
 - Continuous actions: prices
- We try to scale up the model-free methods using approximations for value/policy functions:

$$\hat{v}(s, \mathbf{w}) \approx v_{\pi}(s)$$

$$\text{or } \hat{q}(s, a, \mathbf{w}) \approx q_{\pi}(s, a)$$

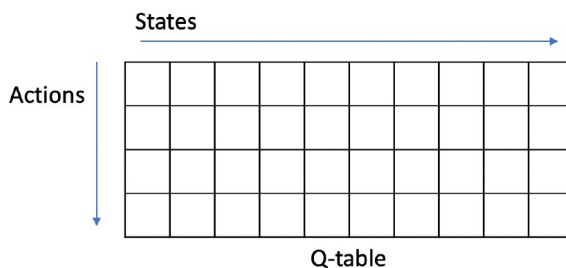
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Value Function Approximation

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- In tabular RL, we represent the value functions by a lookup table:
 - Every state s has an entry $V(s)$
 - Every state-action pair (s,a) has an entry $Q(s,a)$



- Challenges with large MDPs:
 - Too many states or actions to store in memory
 - Too slow to learn the value of each state individually

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Value Function Approximation

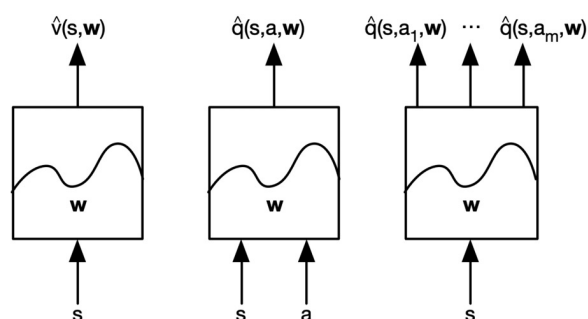
<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-6-value-function-approximation-.pdf>

- Avoid explicitly learning or storing for every single state
 - Dynamics and reward model
 - Value function $V(s)$ and state-action function $Q(s,a)$
 - Policy
- Solution: Estimate approximated value and policy functions:

$$\begin{aligned}\hat{v}(s, \mathbf{w}) &\approx v^\pi(s) \\ \hat{q}(s, a, \mathbf{w}) &\approx q^\pi(s, a) \\ \hat{\pi}(a, s, \mathbf{w}) &\approx \pi(a|s)\end{aligned}$$

- Generalize from seen states to unseen states
- Update the parameter $\mathbf{w}$ using MC or TD learning.

Several Function Designs



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Value Function Approximation

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- Consider differentiable functions as the approximators that can deal with non-stationary and non-iid data:
 - Linear feature representations
 - Neural Nets
- Idea: Use gradient descent to update the value or policy approximators every time we see a sample.
- Value function approximation with an **oracle**.
 - We have the oracle for knowing the true value for $v^\pi(s)$ for any given state s .
 - The objective is to find the best approximate representation of $v^\pi(s)$.
 - Use the MSE with loss function as:

$$J(\mathbf{w}) = \mathbb{E}_\pi \left[(v^\pi(s) - \hat{v}(s, \mathbf{w}))^2 \right]$$

- Then gradient descent:

$$\Delta \mathbf{w} = -\frac{1}{2} \alpha \nabla_{\mathbf{w}} J(\mathbf{w})$$

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \Delta \mathbf{w}$$

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Linear Approximation with an Oracle

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- Represent states using a finite vector with n variables: $\mathbf{x}(s) = \begin{pmatrix} x_1(s) \\ x_2(s) \\ \vdots \\ x_n(s) \end{pmatrix}$
- Represent value function by a linear combination of features: $\hat{V}(s; \mathbf{w}) = \sum_{j=1}^n x_j(s) w_j = \mathbf{x}(s)^T \mathbf{w}$
- Loss function: $J(\mathbf{w}) = \mathbb{E}_\pi [(V^\pi(s) - \hat{V}(s; \mathbf{w}))^2]$
- Stochastic gradient descent to update the weights: $\Delta \mathbf{w} = -\frac{1}{2} \alpha \nabla_{\mathbf{w}} J(\mathbf{w})$
 - Since the function is linear and the loss is MSE:

$$\Delta \mathbf{w} = \alpha (v^\pi(s) - \hat{v}(s, \mathbf{w})) \mathbf{x}(s)$$
 - The loss is convex in $\mathbf{w}$, so SGD will converge to the **global minimum**.
 - With a general approximation function (SGD will only converge to the local minimum):

$$\Delta \mathbf{w} = \alpha (v^\pi(s) - \hat{v}(s, \mathbf{w})) \nabla_{\mathbf{w}} \hat{v}(s, \mathbf{w})$$

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Model Free Evaluation with Approximation

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- In practice, we do not have oracles, just **rewards** collected by **interacting with the environment**.
- But we do obtain the **target** using samples.
- For MC, the target is simply the actual return G_t :

$$\Delta \mathbf{w} = \alpha \left(G_t - \hat{v}(s_t, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{v}(s_t, \mathbf{w})$$

- G_t is an **unbiased estimate** of the oracle $v^\pi(s)$
- MC prediction converges, in both linear and non-linear value function approximation.

Called **semi-gradient**, as we ignore the effect of changing $\mathbf{w}$ on the target.

- For TD(0), the target is the TD target $R_{t+1} + \gamma \hat{v}(s_{t+1}, \mathbf{w})$:

$$\Delta \mathbf{w} = \alpha \left(R_{t+1} + \gamma \hat{v}(s_{t+1}, \mathbf{w}) - \hat{v}(s_t, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{v}(s_t, \mathbf{w})$$

- TD target $R_{t+1} + \gamma \hat{v}(s_{t+1}, \mathbf{w})$ is a biased **estimate** of the oracle $v^\pi(s)$
- Linear TD(0) converges (close) to global minimum.

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Algorithms

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-6-value-function-approximation-pdf>

MC + SGD

```

while (not converged)
   $s_0 \sim p_0, t = 0$ 
  while ( $s_t \neq \text{<term>}$ )
     $a_t \sim \pi(\cdot | s_t)$ 
     $(r_t, s_{t+1}) \sim p(\cdot, \cdot | s_t, a_t)$ 
     $t = t + 1$ 
  end
   $T = t$ 
   $g = \left( V_\phi(s_0) - \sum_{t=0}^{T-1} \gamma^t r_t \right) \nabla_\phi V_\phi(s_0)$ 
  update  $\phi$  using  $g$  with an optimizer
end
  
```

} sample trajectory τ

TD + (semi-)SGD

```

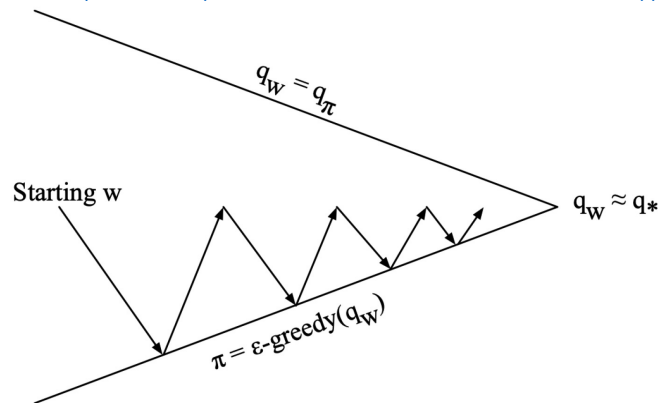
while (not converged)
   $s_0 \sim p_0, a_0 \sim \pi(\cdot | s_0), (r_0, s_1) \sim p(\cdot, \cdot | s_0, a_0)$ 
   $y = \begin{cases} r_0 + \gamma V_\phi(s_1) & \text{if } s_1 \neq \text{<term>} \\ r_0 & \text{if } s_1 = \text{<term>} \end{cases}$ 
   $g = (V_\phi(s_0) - y) \nabla_\phi V_\phi(s_0)$ 
  update  $\phi$  using  $g$  with an optimizer
end
  
```

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Generalized Policy Iteration with Value Function Approximation

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- ① Policy evaluation: **approximate** policy evaluation, $\hat{q}(\cdot, \cdot, \mathbf{w}) \approx q^\pi$
- ② Policy improvement: ϵ -greedy policy improvement

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Model-free Control with Linear Approximation with an Oracle

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- With a similar idea, represent states using a finite vector with n variables: $\mathbf{x}(s, a) = \begin{pmatrix} x_1(s, a) \\ x_2(s, a) \\ \vdots \\ x_n(s, a) \end{pmatrix}$
- Use a linear combination of features to approximate Q-function so as to optimize over actions:

$$\hat{Q}(s, a; \mathbf{w}) = \mathbf{x}(s, a)^T \mathbf{w} = \sum_{j=1}^n x_j(s, a) w_j$$

- Minimize the MSE between the approximate Q-function and the oracle Q-function.

$$J(\mathbf{w}) = \mathbb{E}_\pi[(q^\pi(s, a) - \hat{q}(s, a, \mathbf{w}))^2]$$

- Stochastic gradient descent update:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \nabla_{\mathbf{w}} \mathbb{E}_\pi[(Q^\pi(s, a) - \hat{Q}^\pi(s, a; \mathbf{w}))^2]$$

$$\Delta \mathbf{w} = \alpha (q^\pi(s, a) - \hat{q}(s, a, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s, a, \mathbf{w}) \quad \Delta \mathbf{w} = \alpha (q^\pi(s, a) - \hat{q}(s, a, \mathbf{w})) \mathbf{x}(s, a)$$

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Incremental Control Algorithms

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- In practice, we do not have oracle state-action function $q^\pi(s, a)$, just rewards collected by interacting with the environment.
- For MC, the target is simply the actual return G_t :

$$\Delta \mathbf{w} = \alpha (G_t - \hat{q}(s_t, a_t, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s_t, a_t, \mathbf{w})$$

- For SARSA, the target is the TD target $R_{t+1} + \gamma \hat{q}(s_{t+1}, a_{t+1}, \mathbf{w})$:

$$\Delta \mathbf{w} = \alpha (R_{t+1} + \gamma \hat{q}(s_{t+1}, a_{t+1}, \mathbf{w}) - \hat{q}(s_t, a_t, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s_t, a_t, \mathbf{w})$$

- For Q-learning, the target is TD target $R_{t+1} + \gamma \max_a \hat{q}(s_{t+1}, a, \mathbf{w})$:

$$\Delta \mathbf{w} = \alpha (R_{t+1} + \gamma \max_a \hat{q}(s_{t+1}, a, \mathbf{w}) - \hat{q}(s_t, a_t, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s_t, a_t, \mathbf{w})$$

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Algorithms

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-6-value-function-approximation-pdf>

MC + SGD

```

while (not converged)
   $s_0 \sim p_0, a_0 \sim \pi(\cdot | s_0), t = 0$ 
  while ( $s_t \neq \text{<term>}$ )
     $a_t \sim \pi(\cdot | s_t)$ 
     $(r_t, s_{t+1}) \sim p(\cdot, \cdot | s_t, a_t)$ 
     $t = t + 1$ 
  end
   $T = t$ 
   $g = \left( Q_\phi(s_0, a_0) - \sum_{t=0}^{T-1} \gamma^t r_t \right) \nabla_\phi Q_\phi(s_0, a_0)$ 
  update  $\phi$  using  $g$  with an optimizer
end

```

} sample trajectory τ

TD + (semi-)SGD

Episodic Semi-gradient Sarsa for Estimating $\hat{q} \approx q_*$

Input: a differentiable function $\hat{q} : \mathcal{S} \times \mathcal{A} \times \mathbb{R}^d \rightarrow \mathbb{R}$

Initialize value-function weights $\mathbf{w} \in \mathbb{R}^d$ arbitrarily (e.g., $\mathbf{w} = \mathbf{0}$)

Repeat (for each episode):

$S, A \leftarrow$ initial state and action of episode (e.g., ϵ -greedy)

Repeat (for each step of episode):

Take action A , observe R, S'

If S' is terminal:

$\mathbf{w} \leftarrow \mathbf{w} + \alpha [R - \hat{q}(S, A, \mathbf{w})] \nabla \hat{q}(S, A, \mathbf{w})$

Go to next episode

Choose A' as a function of $\hat{q}(S', \cdot, \mathbf{w})$ (e.g., ϵ -greedy)

$\mathbf{w} \leftarrow \mathbf{w} + \alpha [R + \gamma \hat{q}(S', A', \mathbf{w}) - \hat{q}(S, A, \mathbf{w})] \nabla \hat{q}(S, A, \mathbf{w})$

$S \leftarrow S'$

$A \leftarrow A'$

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Convergence of Control Algorithms

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- SARSA: $\Delta \mathbf{w} = \alpha (R_{t+1} + \gamma \hat{q}(s_{t+1}, a_{t+1}, \mathbf{w}) - \hat{q}(s_t, a_t, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s_t, a_t, \mathbf{w})$
- Q-learning: $\Delta \mathbf{w} = \alpha (R_{t+1} + \gamma \max_a \hat{q}(s_{t+1}, a, \mathbf{w}) - \hat{q}(s_t, a_t, \mathbf{w})) \nabla_{\mathbf{w}} \hat{q}(s_t, a_t, \mathbf{w})$
- TD w. value function approximation does **NOT** follow the gradient of any loss function.
- The updates essentially involves **approximate Bellman backup + fitting the underlying value function**.
- TD may diverge in **off-policy (e.g., Q-learning)** or **non-linear approximation** settings.
- Challenge for off-policy control: Behavior policy and target policy are not identical, so value function approximation may diverge.

Convergence of Approximate Model-free Control

Algorithm	Table Lookup	Linear	Non-Linear
Monte-Carlo Control	✓	(✓)	X
Sarsa	✓	(✓)	X
Q-Learning	✓	X	X

(✓) moves around the near-optimal value function

Deadly Triad:

- Bootstrapping
- Function approximation
- Off-policy learning

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Agenda

- Value Function Approximation
- **DQN**
- Policy-based Methods
- Modern DRL
- RL for LLM

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Going Beyond Linear

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

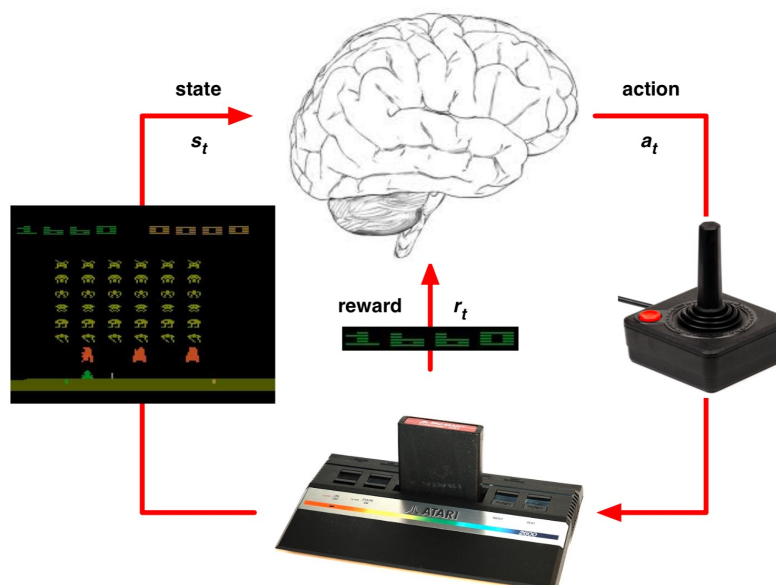
- So far, we have focused on **linear approximators**.
 - Linear approximations work well given the right set of features.
 - It requires **manual design** of the feature set.
- We know that DNN is a much better representation tool if we have a large data set.
- Question: How can we leverage **deep learning** for function approximation and model-free control of MDPs?

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Deep Reinforcement Learning in Atari

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>



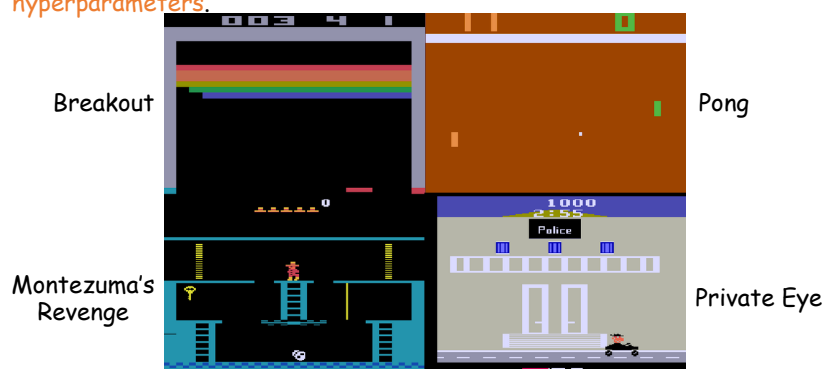
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Deep Q-Network (DQN) for Atari

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- Groud-breaking paper by DeepMind: [Human-level control through deep reinforcement learning](#)
V Mnih, K Kavukcuoglu, D Silver, AA Rusu, J Veness... - nature, 2015 - nature.com
... large neural networks using a reinforcement learning signal and stochastic gradient descent in a stable manner— illustrated by the temporal evolution of two indices of learning (the ...
☆ 保存 勿 引用 被引用次数: 40192 相关文章 所有 55 个版本
- DQN represents the action-value function with DNN approximator.
- DQN reaches a professional human gaming level across many Atari games using the same network and hyperparameters.



Game	Linear	Deep Network
Breakout	3	3
Enduro	62	29
River Raid	2345	1453
Seaquest	656	275
Space Invaders	301	302

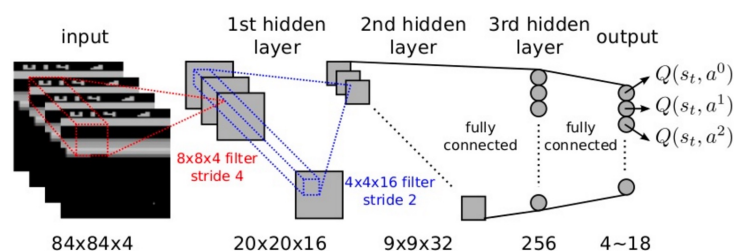
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Deep Q-Network (DQN) for Atari

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- End-to-end learning of $Q(s, a)$ from input fixel frame.
- Input state s is a stack of raw pixels from the latest 4 frames.
- Output of $Q(s, a)$ is 18 buttons.
- Reward is the change in score for that step.
- Network arthitecture and hyperparameters fixed across all games:



Abstract

The theory of reinforcement learning provides a normative account¹, deeply rooted in psychological² and neuroscientific³ perspectives on animal behaviour, of how agents may optimize their control of an environment. To use reinforcement learning successfully in situations approaching real-world complexity, however, agents are confronted with a difficult task: they must derive efficient representations of the environment from high-dimensional sensory inputs, and use these to generalize past experience to new situations. Remarkably, humans and other animals seem to solve this problem through a harmonious combination of reinforcement learning and hierarchical sensory processing systems^{4,5}, the former evidenced by a wealth of neural data revealing notable parallels between the phasic signals emitted by dopaminergic neurons and temporal difference reinforcement learning algorithms⁶. While reinforcement learning agents have achieved some successes in a variety of domains^{6,7,8}, their applicability has previously been limited to domains in which useful features can be handcrafted, or to domains with fully observed, low-dimensional state spaces. Here we use recent advances in training deep neural networks^{9,10,11} to develop a novel artificial agent, termed a deep Q-network, that can learn successful policies directly from high-dimensional sensory inputs using end-to-end reinforcement learning. We tested this agent on the challenging domain of classic Atari 2600 games¹². We demonstrate that the deep Q-network agent, receiving only the pixels and the game score as inputs, was able to surpass the performance of all previous algorithms and achieve a level comparable to that of a professional human games tester across a set of 49 games, using the same algorithm, network architecture and hyperparameters. This work bridges the divide between high-dimensional sensory inputs and actions, resulting in the first artificial agent that is capable of learning to excel at a diverse array of challenging tasks.

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Deep Reinforcement Learning

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- We leverage DNNs to represent:
 - Value functions (Q & V)
 - Policy functions (π)
 - World model
- Loss function is optimized by stochastic gradient descent (SGD)
- Core challenges:
 - Data inefficiency: Too many model parameters to optimize.
 - Deadly Triad creates instability and divergence in training:
 - Nonlinear function approximation
 - Bootstrapping
 - Off-policy training
- Deep Q-learning (DQN) addresses **correlations between samples** and **non-stationary targets** through:
 - **Experience replay**: Sample experience randomly from data to update weights, **reducing correlations** between data points.
 - **Fixed Q-targets**: Fix the target networks' weights while updating, **improving stability**.

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DQN: Experience Replay

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- To **reduce the correlations** among samples, store the transition (s_t, a_t, r_t, s_{t+1}) in **replay memory** D .

s_1, a_1, r_1, s_2	$\rightarrow s, a, r, s'$
s_2, a_2, r_2, s_3	
s_3, a_3, r_3, s_4	
...	
s_t, a_t, r_t, s_{t+1}	

- To perform experience replay, repeat the following:
 1. Sample an experience tuple from the dataset: $(s, a, r, s') \sim D$.
 2. Compute the target value for the sampled tuple: $r + \gamma \max_{a'} \hat{Q}(s', a', \mathbf{w})$
 3. Use SGD to update the network weights $\mathbf{w}$:

$$\Delta \mathbf{w} = \alpha \left(r + \gamma \max_{a'} \hat{Q}(s', a', \mathbf{w}) - Q(s, a, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{Q}(s, a, \mathbf{w})$$

- Use target as a scalar, but network weights will get updated on the next round, changing the target value.

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DQN: Fixed Targets

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- To **improve stability**, fix the target weights used in the target calculation for multiple updates.
- Target network uses a different set of weights than the weights being updated.
- Let a different set of parameter $\mathbf{w}^-$, be the set of weights used in the target, and $\mathbf{w}$ be the weights that are being updated.
- To perform experience replay with fixed target, repeat the following:
 1. Sample an experience tuple from the dataset: $(s, a, r, s') \sim D$
 2. Compute the target value for the sampled tuple: $r + \gamma \max_{a'} \hat{Q}(s', a', \mathbf{w}^-)$
 3. Use stochastic gradient descent to update the network weights:

$$\Delta \mathbf{w} = \alpha \left(r + \gamma \max_{a'} \hat{Q}(s', a', \mathbf{w}^-) - Q(s, a, \mathbf{w}) \right) \nabla_{\mathbf{w}} \hat{Q}(s, a, \mathbf{w})$$

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DQN Algorithm

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

```

1: Input  $\mathbf{C}, \alpha, D = \{\}$ , Initialize  $\mathbf{w}, \mathbf{w}^- = \mathbf{w}, t = 0$ 
2: Get initial state  $s_0$ 
3: loop
4:   Sample action  $a_t$  given  $\epsilon$ -greedy policy for current  $\hat{Q}(s_t, a; \mathbf{w})$ 
5:   Observe reward  $r_t$  and next state  $s_{t+1}$ 
6:   Store transition  $(s_t, a_t, r_t, s_{t+1})$  in replay buffer  $D$ 
7:   Sample random minibatch of tuples  $(s_i, a_i, r_i, s_{i+1})$  from  $D$ 
8:   for  $j$  in minibatch do
9:     if episode terminated at step  $i + 1$  then
10:       $y_i = r_i$ 
11:     else
12:       $y_i = r_i + \gamma \max_{a'} \hat{Q}(s_{i+1}, a'; \mathbf{w}^-)$ 
13:     end if
14:     Do gradient descent step on  $(y_i - \hat{Q}(s_i, a_i; \mathbf{w}))^2$  for parameters  $\mathbf{w}$ :  $\Delta \mathbf{w} = \alpha (y_i - \hat{Q}(s_i, a_i; \mathbf{w})) \nabla_{\mathbf{w}} \hat{Q}(s_i, a_i; \mathbf{w})$ 
15:   end for
16:    $t = t + 1$ 
17:   if  $\text{mod}(t, \mathbf{C}) == 0$  then
18:      $\mathbf{w}^- \leftarrow \mathbf{w}$ 
19:   end if
20: end loop

```

Sample code:

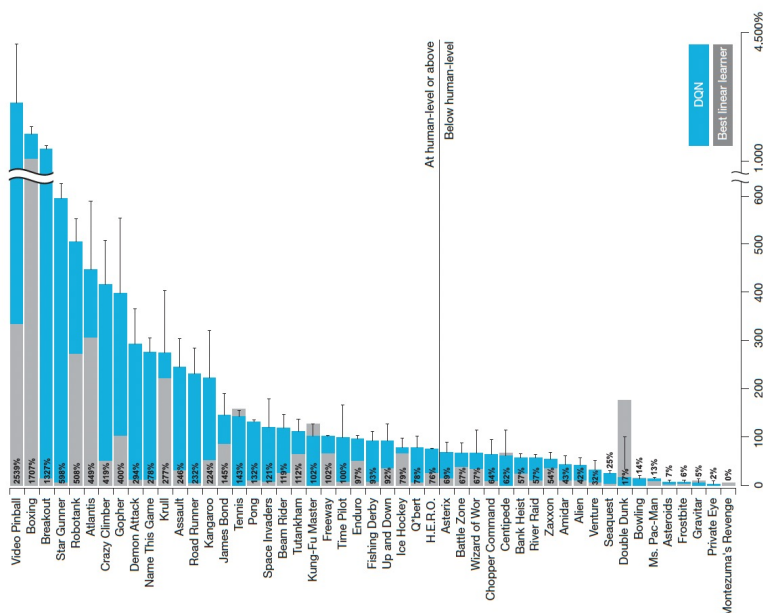
- https://www.tensorflow.org/agents/tutorials/0_intro_rl
- https://keras.io/examples/rl/deep_q_network_breakout/

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Performance of DQN on Atari

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>



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Performance of DQN on Atari

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

	Replay Fixed-Q	Replay Q-learning	No replay Fixed-Q	No replay Q-learning
Breakout	316.81	240.73	10.16	3.17
Enduro	1006.3	831.25	141.89	29.1
River Raid	7446.62	4102.81	2867.66	1453.02
Seaquest	2894.4	822.55	1003	275.81
Space Invaders	1088.94	826.33	373.22	301.99

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Why Fixed Targets?

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- In the original Q-learning with value function approximation, both Q estimation and Q target shifts at each time step.
- Think of a cat (Q estimation) is chasing a mouse (Q target), with the purpose that the cat reduces the distance to the mouse.



- If both the cat and the mouse are moving, we will observe an **oscillated training history**; so, we fix the target for a period of time during training.



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Extensions to DQN

<https://web.stanford.edu/class/cs234/slides/lecture4pre.pdf>

- The success of DQN on Atari inspires a set of papers improving deep reinforcement learning.
 - Double DQN** (AAAI 2016), **Prioritized Replay** (ICLR 2016), **Dueling DQN** (best paper at ICML 2016)
 - Also, huge economic values through self-driving cars.

Deep reinforcement learning with double q-learning

H Van Hasselt, A Guez, D Silver - ... of the AAAI conference on artificial ..., 2016 - ojs.aaai.org

... The popular Q-learning algorithm is known to overestimate ... algorithm, which combines

Q-learning with a deep neural network, ... We then show that the idea behind the Double Q-learning ...

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Prioritized experience replay

T Schaul, J Quan, I Antonoglou, D Silver - arXiv preprint arXiv:1511.05952, 2015 - arxiv.org

... To demonstrate the potential effectiveness of prioritizing replay by TD ... 'prioritization' algorithm.

This algorithm stores the last encountered TD error along with each transition in the replay ...

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Dueling network architectures for deep reinforcement learning

Z Wang, T Schaul, M Hesse!... - ... machine learning, 2016 - proceedings.mlr.press

... Still, many of these applications use conventional architectures, such as convolutional ...

neural network architecture for model-free reinforcement learning. Our dueling network represents ...

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Deep Q-Learning for Dynamic Coupon Targeting

informs

https://pubsonline.informs.org/journal/mksc

MARKETING SCIENCE

Articles in Advance, pp. 1–22
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Dynamic Coupon Targeting Using Batch Deep Reinforcement Learning: An Application to Livestream Shopping

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Abstract. We present an empirical framework for creating dynamic coupon targeting strategies for high-dimensional and high-frequency settings, and we test its performance using a large-scale field experiment. The framework captures consumers' intertemporal tradeoffs associated with dynamic pricing and does not rely on functional form assumptions about consumers' decision-making processes. The model is estimated using batch deep reinforcement learning (BDRL), which relies on Q-learning, a model-free solution that can mitigate model bias. It leverages deep neural networks to represent the high-dimensional state space and alleviate the curse of dimensionality. The empirical application is in a multibillion-dollar livestream shopping contest. Our BDRL solution increases the platform's revenue by twice as much as static targeting policies and by 20% more than the model-based solution. The comparative advantage of BDRL comes from more effective and automatic targeting of consumers based on both heterogeneity and dynamics, using exceptionally rich, nuanced differences among consumers and across time. We find that price skimming, reducing discounts for attractive hosts, and increasing the coupon discount level at a faster rate for low spenders are effective strategies based on dynamics, consumer heterogeneity, and the two combined, respectively.

History: K. Sudhir served as the senior editor and John Hauser served as associate editor for this article. **Funding:** Partial financial support was received from the NYU Center for Global Economy and Business. **Supplemental Material:** The data files and online appendices are available at <https://doi.org/10.1287/mksc.2022.1403>.

Keywords: dynamic pricing • coupon • deep reinforcement learning • reference price • livestream shopping • targeting

Dynamic coupon targeting using batch deep reinforcement learning: An application to livestream shopping

X. Liu • Marketing Science, 2023 • pubsonline.informs.org

... incorporates the intertemporal tradeoffs in dynamic coupon targeting to design a policy ...

dynamic coupon targeting policies? (3) What are the gains, if any, from using a dynamic targeting ...

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Algorithm 2 (BCQ)

1. Input: Batch data $\mathcal{B}$, horizon T , target network update rate Υ , mini-batch size N , threshold τ .
2. Initialize Q-networks Q_θ , generative model G_ω (trained in a standard supervised learning fashion, with cross-entropy loss), and target networks $Q_{\theta'}$, with $\theta' \leftarrow \theta$.
3. for $t = 1$ to T , do:
4. Sample mini-batch M of N transitions (S, A, R, S') from $\mathcal{B}$.
5. $A' = \arg \max_{A'} \left| \frac{G_\omega(A'|S')}{\max_A G_\omega(A|S')} \right| > \tau$ $Q_\theta(S', A')$
6. $\theta \leftarrow \arg \min_\theta \sum_{(S, A, R, S') \in M} (R + \gamma Q_{\theta'}(S', A') - Q_\theta(S, A))$
7. $\omega \leftarrow \arg \min_\omega - \sum_{(S, A) \in M} \log G_\omega(A | S)$
8. If $t \bmod \Upsilon = 0$: $\theta' \leftarrow \theta$
9. end for

Table 7. Model Comparison Based on the Doubly Robust Estimator

		1. Static policy with homogeneous treatments		2. Static policy with heterogeneous treatments			3. Model-based dynamic policy with heterogeneous treatments		4. Model-free dynamic policy with heterogeneous treatments
		A: Regression		B: GBDT	C: DNN	D: ORF	E: Structural ^a		F: Proposed BDRL
CLV	Mean	6.53		7.52	6.95	7.49	8.37		9.53
(Return)	Std	2.00		2.29	2.27	2.29	2.77		3.23
Gain	Mean	11%		28%	18%	28%	43%		62%
t test ^b		222.99		524.25	346.60	515.47	714.27		945.57
p value		<0.001		<0.001	<0.001	<0.001	<0.001		<0.001

Notes. All the gains are compared with the mean CLV of ¥5.87. The total sample size is 1,020,898 consumers.

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DRL for Ensembling Experimentation

informs

https://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE

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Ensemble Experiments to Optimize Interventions Along the Customer Journey: A Reinforcement Learning Approach

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Abstract. Firms adopt randomized experiments to evaluate various interventions (e.g., website design, creative content, and pricing). However, most randomized experiments are designed to identify the impact of one specific intervention. The literature on randomized experiments lacks a holistic approach to optimize a sequence of interventions along the customer journey. Specifically, locally optimal interventions unveiled by randomized experiments might be globally suboptimal when considering their interdependence as well as the long-term rewards. Fortunately, the accumulation of a large number of historical experiments creates exogenous interventions at different stages along the customer journey and provides a new opportunity. This study integrates multiple experiments within the reinforcement learning (RL) framework to tackle the questions that cannot be answered by stand-alone randomized experiments. How can we learn optimal policy with a sequence of interventions along the customer journey based on an ensemble of historical experiments? Additionally, how can we learn from multiple historical experiments to guide future intervention trials? We propose a Bayesian recurrent Q-network model that leverages the exogenous interventions from multiple experiments to learn their effectiveness at different stages of the customer journey and optimize them for long-term rewards. Beyond optimization within the existing interventions, the Bayesian model also estimates the distribution of rewards, which can guide subject allocation in the design of future experiments to optimally balance exploration and exploitation. In summary, the proposed model creates a two-way complementarity between RL and randomized experiments, and thus, it provides a holistic approach to learning and optimizing interventions along the customer journey.

History: Accepted by Anindya Ghose, information systems. **Funding:** This work was supported by Adobe Faculty Research Award and the Marketing Science Institute Research Grant. **Supplemental Material:** The data files and online appendices are available at <https://doi.org/10.1287/mnsc.2023.4914>.

Keywords: reinforcement learning • customer journey • long-term reward optimization • Bayesian recurrent Q-network model (BRQN) • randomized experiment • experiment design

Ensemble experiments to optimize interventions along the customer journey: A reinforcement learning approach

Y. Song, T. Sun • Management Science, 2024 • pubsonline.informs.org

... of interventions along the customer journey based on an ensemble of historical experiments ...

we learn from multiple historical experiments to guide future intervention trials? We propose ...

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Algorithm 1 (Bayesian Recurrent Q-Network Estimation)

1. Set the empty reply buffer $RB()$, and load the historical experiment data into the buffer.
2. Initialize RNN parameter θ and Bayesian parameter w_a for each action.
3. Set N as the interval for Bayesian sampling and M as the interval for target model update.
4. Set $\theta_{\text{target}} = \theta$ and $w_{a, \text{target}} = w_a$.
5. **while** $\text{step} \leq \text{num_steps}$ **do**
6. Load a minibatch from $RB()$ with probability relative to TD error in Equation (8)
7. Get s_t based on RNN
8. Get s_{t+1} based on RNN_{target}
9. Get TD between state s_t and s_{t+1} based on Equation (8)
10. Update the RNN parameter θ based on Equation (9)
11. Bayesian posterior update based on Equations (3) and (4)
12. For every N steps: Thompson sampling $w_a \sim N(\mu, \Sigma)$
13. For every M steps: $\theta_{\text{target}} = \theta$ and $w_{a, \text{target}} = w_a$
14. $\text{step} = \text{step} + 1$
15. **end while**

Table 5. Average Reward per Episode (Cents) of BRQN with Different Experimental Data

	Fixed-M PERS @ 1	Fixed-M PERS @ 4	Fixed-M PERS @ 8
1 Experiment	502.4	2,027.9	4,190.2
5 Experiments	515.1	2,167.5	4,711.8
10 Experiments	536.5	2,429.3	5,708.6

Notes. We train the BRQN model with data from only 1 experiment (the one with the maximal number of treated users), 5 experiments (the top five with the greatest number of treated users), and all 10 experiments. Then, we adopt the fixed-M PERS to sample episodes with lengths of one, four, and eight. The average reward of the sampled episode across different settings shows promise of fueling RL with multiple experiments.

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RL for Business Decision Making



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Deep Reinforcement Learning for Sequential Targeting

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Abstract. Deep reinforcement learning (DRL) has opened up many unprecedented opportunities in revolutionizing the digital marketing field. In this study, we designed a DRL-based personalized targeting strategy in a sequential setting. We show that the strategy is able to address three important challenges of sequential targeting: (1) forward looking (balancing between a firm's current revenue and future revenues), (2) earning while learning (maximizing profits while continuously learning through exploration-exploitation), and (3) scalability (coping with a high-dimensional state and policy space). We illustrate this through a novel design of a DRL-based artificial intelligence (AI) agent. To better adapt DRL to complex consumer behavior dimensions, we proposed a quantization-based uncertainty learning heuristic for efficient exploration-exploitation. Our policy evaluation results through simulation suggest that the proposed DRL agent generates 26.75% more long-term revenues than can the non-DRL approaches on average and learns 76.92% faster than the second fastest model among all benchmarks. Further, in order to better understand the potential underlying mechanisms, we conducted multiple interpretability analyses to explain the patterns of learned optimal policy at both the individual and population levels. Our findings provide important managerial-relevant and theory-consistent insights. For instance, consecutive price promotions at the beginning can capture price-sensitive consumers' immediate attention, whereas carefully spaced nonpromotional "cooldown" periods between price promotions can allow consumers to adjust their reference points. Additionally, consideration of future revenues is necessary from a long-term horizon, but weighing the future too much can also dampen revenues. In addition, analyses of heterogeneous treatment effects suggest that the optimal promotion sequence pattern highly varies across the consumer engagement stages. Overall, our study results demonstrate DRL's potential to optimize these strategies' combination to maximize long-term revenues.

History: Accepted by Kartik Hosanagar, information systems.

Supplemental Material: The data files and online appendix are available at <https://doi.org/10.1287/mnsc.2022.4621>.

Keywords: deep reinforcement learning • DRL • sequential targeting • promotions • forward looking • exploration-exploitation • scalability • AI

Deep reinforcement learning for sequential targeting

W Wang, B Li, X Luo, X Wang - Management Science, 2023 - pubsonline.informs.org

... based personalized targeting strategy in a sequential setting. ... three important challenges of sequential targeting: (1) ... while learning (maximizing profits while continuously learning ...

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Demand-Aware Career Path Recommendations: A Reinforcement Learning Approach

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Abstract. A skill's value depends on dynamic market conditions. To remain marketable, contractors need to keep reskilling themselves continuously. But choosing new skills to learn is an inherently hard task. Contractors have very little information about current and future market conditions, which often results in poor learning choices. Recommendation frameworks could reduce uncertainty in learning choices. However, conventional approaches would likely be inefficient; they would model previous (often poor) observed contractor learning behaviors to provide future career path recommendations while ignoring current market trends. This work proposes a framework that combines reinforcement learning, Bayesian inference, and gradient boosting to provide recommendations on how contractors should behave when choosing new skills to learn. Compared with standard recommender systems, this framework does not learn from previous (often poor) behaviors to make future recommendations. Instead, it relies on a Markov decision process to operate on a graph of feasible actions and dynamically recommend profitable career paths. The framework uses market information to identify current trends and project future wages. Based on this information, it recommends feasible, relevant actions that a contractor can take to learn new, in-demand skills. Evaluation of the framework on 1.73 million job applications from an online labor market shows that its implementation could increase (1) the marketplace's revenue by up to 6%, (2) contractors' wages by 22%, and (3) the diversity of new skill acquisitions by 47%. A comparison with alternative recommender systems highlights the limitations of approaches that make recommendations based on previously observed learning behaviors.

History: Accepted by Chris Forman, information systems.

Supplemental Material: The online appendixes are available at <https://doi.org/10.1287/mnsc.2020.3727>.

Keywords: skill recommendations • artificial intelligence • online labor markets • Markov decision process • information systems: enabling technologies

Demand-aware career path recommendations: A reinforcement learning approach

M Kokkodis, PG Ipeirotis - Management science, 2021 - pubsonline.informs.org

... Recommendation frameworks could reduce uncertainty in ... to provide future career path recommendations while ignoring ... boosting to provide recommendations on how contractors .

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Agenda

- Value Function Approximation
- DQN
- Policy-based Methods
- Modern DRL
- RL for LLM

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From Value to Policy Approximations

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- So far, we have focused on approximating the value functions $v^\pi(s)$ and $q^\pi(s, a)$ given the policy π .
- Another perspective is to parameterize the policy function $\pi_\theta(a|s)$.

- Question: Solving the following policy optimization problem

$$\underset{\theta}{\text{maximize}} \quad \mathcal{J}(\theta) = \mathbb{E}_{s_0 \sim p_0} [V^{\pi_\theta}(s_0)] = \mathbb{E}_{s_0 \sim p_0}^\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$$

where π_θ is represented by a DNN with parameter θ .

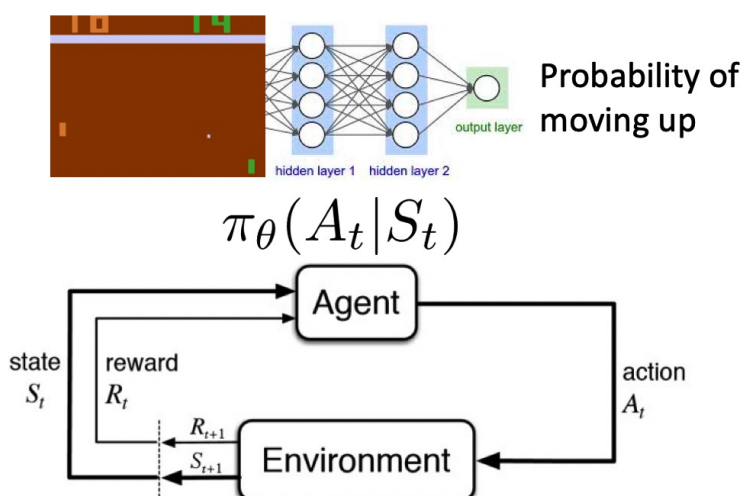
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Policy Optimization

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- The action and reward from the policy is all we need to optimize the policy directly.



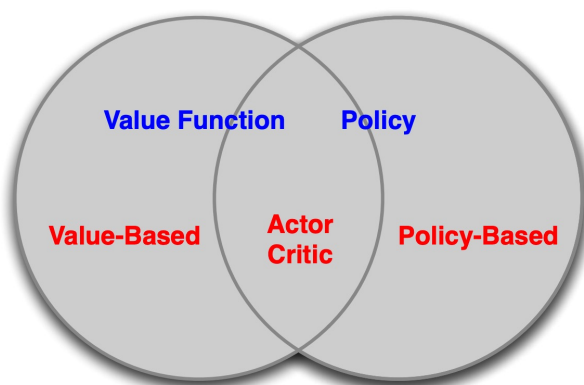
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Value-based RL vs. Policy-based RL

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- **Value-based RL** learns the value function and optimizes the policy based on the learned value function.
- **Policy-based RL** learns the optimal policy directly without obtaining the value function.
- **Actor-critic** learns both policy and value functions.



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Pros and Cons of Policy-based RL

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- We can directly learn **stochastic policies** (important for games).
- Policy-based RL is effective when action spaces are **high-dimensional or continuous** (e.g., LLMs).
- Better **convergency** properties, similar to policy iteration.
- Typically **high variance**.
- Typically convergence to **local optimum**.
 - The entire policy space is harder to optimize.

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Policy Gradient for One-Step MDPs

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- Consider an MDP with **one-step**.
 - Start with $s \sim d(s)$
 - Terminate after one step with reward $r = R(s, a)$

- The reward of policy $\pi_\theta(s, a)$: $J(\theta) = \mathbb{E}_{\pi_\theta}[r]$

$$= \sum_{s \in \mathcal{S}} d(s) \sum_{a \in \mathcal{A}} \pi_\theta(s, a) r$$

- The gradient is:

$$\begin{aligned} \nabla_\theta J(\theta) &= \sum_{s \in \mathcal{S}} d(s) \sum_{a \in \mathcal{A}} \pi_\theta(s, a) \nabla_\theta \log \pi_\theta(s, a) r \\ &= \mathbb{E}_{\pi_\theta}[r \nabla_\theta \log \pi_\theta(s, a)] \end{aligned} \quad \begin{aligned} \nabla_\theta \pi_\theta(s, a) &= \pi_\theta(s, a) \frac{\nabla_\theta \pi_\theta(s, a)}{\pi_\theta(s, a)} \\ &= \pi_\theta(s, a) \nabla_\theta \log \pi_\theta(s, a) \end{aligned}$$

Score Function or Gradient of
(Log-)Likelihood Ratio

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Policy Gradient for Multi-Step MDPs

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- Consider a **state-action trajectory** from one episode as:

$$\tau = (s_0, a_0, r_1, \dots, s_{T-1}, a_{T-1}, r_T, s_T) \sim (\pi_\theta, P(s_{t+1}|s_t, a_t))$$

- Define $R(\tau) = \sum_{t=0}^T R(s_t, a_t)$ as the sum of rewards for a trajectory τ (assuming $\gamma = 1$).

- Policy value is:

$$J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{T-1} R(s_t, a_t) \right] = \sum_{\tau} P(\tau; \theta) R(\tau)$$

where $P(\tau; \theta) = \mu(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t)$ is the probability over the trajectory τ if policy π_θ is played.

- Our goal is to find the optimal policy parameter θ^* .

$$\theta^* = \arg \max_{\theta} J(\theta) = \arg \max_{\theta} \sum_{\tau} P(\tau; \theta) R(\tau)$$

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Taking Gradient

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

$$\theta^* = \arg \max_{\theta} J(\theta) = \arg \max_{\theta} \sum_{\tau} P(\tau; \theta) R(\tau)$$

- Take the gradient w.r.t. θ :

$$\begin{aligned} \nabla_{\theta} J(\theta) &= \nabla_{\theta} \sum_{\tau} P(\tau; \theta) R(\tau) \\ &= \sum_{\tau} \nabla_{\theta} P(\tau; \theta) R(\tau) \\ &= \sum_{\tau} \frac{P(\tau; \theta)}{P(\tau; \theta)} \nabla_{\theta} P(\tau; \theta) R(\tau) \\ &= \sum_{\tau} P(\tau; \theta) R(\tau) \frac{\nabla_{\theta} P(\tau; \theta)}{P(\tau; \theta)} \\ &= \sum_{\tau} P(\tau; \theta) R(\tau) \nabla_{\theta} \log P(\tau; \theta) \end{aligned}$$

- Decompose $\nabla_{\theta} \log P(\tau; \theta)$:

$$\begin{aligned} \nabla_{\theta} \log P(\tau; \theta) &= \nabla_{\theta} \log \left[\mu(s_0) \prod_{t=0}^{T-1} \pi_{\theta}(a_t | s_t) p(s_{t+1} | s_t, a_t) \right] \\ &= \nabla_{\theta} \left[\log \mu(s_0) + \sum_{t=0}^{T-1} \log \pi_{\theta}(a_t | s_t) + \log p(s_{t+1} | s_t, a_t) \right] \\ &= \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \end{aligned}$$

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Estimate the Gradient and REINFORCE

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

$$\nabla_{\theta} J(\theta) = E_{\tau \sim p_{\theta}(\tau)} \left[\left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) \right) \left(\sum_{t=1}^T r(\mathbf{s}_t, \mathbf{a}_t) \right) \right]$$

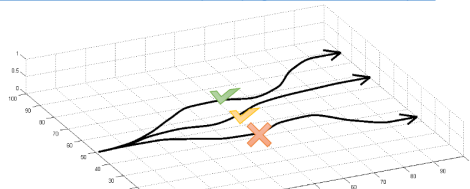
$$\nabla_{\theta} J(\theta) \approx \frac{1}{N} \sum_{i=1}^N \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t}) \right) \left(\sum_{t=1}^T r(\mathbf{s}_{i,t}, \mathbf{a}_{i,t}) \right)$$

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$$

Full algorithm:

1. sample $\{\tau^i\}$ from $\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$
2. $\nabla_{\theta} J(\theta) \approx \sum_i \left(\sum_t \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t^i | \mathbf{s}_t^i) \right) \left(\sum_t r(\mathbf{s}_t^i, \mathbf{a}_t^i) \right)$
3. $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$

“REINFORCE algorithm”, vanilla policy gradient + Monte-Carlo



Run policy to collect batch of data

Improve policy using batch of data

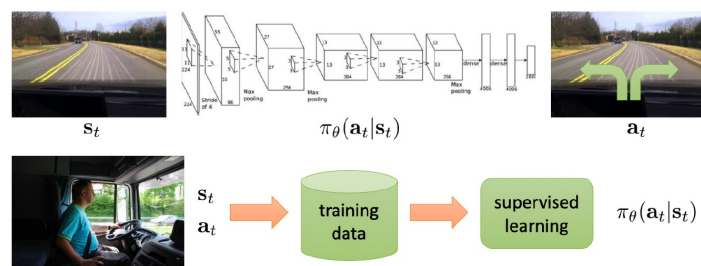
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Policy Gradient vs. Maximum Likelihood

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

- ① Policy gradient estimator: $\nabla_{\theta} J(\theta) \approx \frac{1}{M} \sum_{m=1}^M \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_{t,m} | s_{t,m}) \right) \left(\sum_{t=1}^T r(s_{t,m}, a_{t,m}) \right)$
- ② Maximum likelihood estimator: (Also called imitation learning or behavioral cloning.)
 $\nabla_{\theta} J_{ML}(\theta) \approx \frac{1}{M} \sum_{m=1}^M \left(\sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(a_{t,m} | s_{t,m}) \right)$
- ③ Interpretation: **good action is made more likely, bad action is made less likely**



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Policy Gradient Theorem

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- The **policy gradient theorem** generalizes the likelihood ratio approach:

Theorem

For any differentiable policy $\pi_{\theta}(s, a)$,
 for any of the policy objective function $J = J_1$, (episodic reward), J_{avR} (average reward per time step), or $\frac{1}{1-\gamma} J_{avV}$ (average value),
 the policy gradient is

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) Q^{\pi_{\theta}}(s, a)]$$

- Basically, the policy gradient theorem says **policy gradient = Q-function following the policy weighted by the score function of the policy**.

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Score Function

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>

- **Discrete action policy:** policy function is softmax, and the score function is as follows:

$$\pi_{\theta}(s, a) = \frac{\exp(\phi(s, a)^T \theta)}{\sum_{a'} \exp(\phi(s, a')^T \theta)} \quad \nabla_{\theta} \log \pi_{\theta}(s, a) = \phi(s, a) - \mathbb{E}_{\pi_{\theta}}[\phi(s, \cdot)]$$

- **Continuous action policy:** policy function is Gaussian with mean linear in states, and the score function is as follows:

$$\begin{aligned} \mu(s) &= \phi(s)^T \theta \\ a &\sim \mathcal{N}(\mu(s), \sigma^2) \end{aligned} \quad \nabla_{\theta} \log \pi_{\theta}(s, a) = \frac{(a - \mu(s))\phi(s)}{\sigma^2}$$

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Large Variance of Policy Gradient

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

- We have the following **approximate gradient estimate**:

$$\nabla_{\theta} J(\theta) \approx \frac{1}{m} \sum_{i=1}^m R(\tau_i) \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t^i | s_t^i)$$

- This is **unbiased** but **very noisy**.
- Two fixes:
 - Leverage **temporal causality**.
 - Subtract a **baseline**.

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Temporal Causality

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

- Temporal causality: policy at time t' cannot affect reward at time t for $t < t'$

$$\begin{aligned}\nabla_{\theta} \mathbb{E}_{\tau} [r_{t'}] &= \mathbb{E}_{\tau} \left[r_{t'} \sum_{t=0}^{t'} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \\ \nabla_{\theta} J(\theta) &= \nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R] = \mathbb{E}_{\tau} \left[\sum_{t'=0}^{T-1} r_{t'} \sum_{t=0}^{t'} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \\ &= \mathbb{E}_{\tau} \left[\sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \sum_{\substack{t'=t \\ t'=t}}^{T-1} r_{t'} \right] \\ &= \mathbb{E}_{\tau} \left[\sum_{t=0}^{T-1} G_t \cdot \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]\end{aligned}$$

We have the following estimated gradient update: $\nabla_{\theta} \mathbb{E}[R] \approx \frac{1}{m} \sum_{i=1}^m \sum_{t=0}^{T-1} G_t^{(i)} \cdot \nabla_{\theta} \log \pi_{\theta}(a_t^i | s_t^i)$

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Subtracting Baseline

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

$$\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R] = \mathbb{E}_{\tau} \left[\sum_{t=0}^{T-1} \mathbf{G}_t \cdot \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right]$$

- $G_t = \sum_{t'=t}^{T-1} r_{t'}$ is the return for a trajectory which might have **high variance**.

- We can **subtract a constant baseline** $b(s_t) = \mathbb{E}[r_t + r_{t+1} + \dots + r_{T-1}]$

$$\nabla_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R] = \mathbb{E}_{\tau} \left[\sum_{t=0}^{T-1} (G_t - b(s_t)) \cdot \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) \right] \quad \begin{array}{l} \text{Increase the log-prob of the action proportional} \\ \text{to how much returns are better than expectation.} \end{array}$$

$$\mathbb{E}_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t) \right] = 0,$$

$$E_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b(s_t)) \right] = E_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t \right]$$

Subtracting the baseline is **unbiased** and **reduces variance**.

$$\text{Var}_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) (G_t - b(s_t)) \right] < \text{Var}_{\tau} \left[\nabla_{\theta} \log \pi_{\theta}(a_t | s_t) G_t \right]$$

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Vanilla Policy Gradient with Baseline

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

procedure POLICY GRADIENT(α)

Initialize policy parameters θ and baseline values $b(s)$ for all s , e.g. to 0

for iteration = 1, 2, ... **do**

Collect a set of m trajectories by executing the current policy π_θ

for each time step t of each trajectory $\tau^{(i)}$ **do**

Compute the *return* $G_t^{(i)} = \sum_{t'=t}^{T-1} r_{t'}$

Compute the *advantage estimate* $\hat{A}_t^{(i)} = G_t^{(i)} - b(s_t)$

Re-fit the baseline to the empirical returns by updating $\mathbf{w}$ to minimize

$$\sum_{i=1}^m \sum_{t=0}^{T-1} \|b(s_t) - G_t^{(i)}\|^2$$

Update policy parameters θ using the policy gradient estimate $\hat{g}$

$$\hat{g} = \sum_{i=1}^m \sum_{t=0}^{T-1} \hat{A}_t^{(i)} \nabla_\theta \log \pi_\theta(a_t^{(i)} | s_t^{(i)})$$

with an optimizer like SGD ($\theta \leftarrow \theta + \alpha \cdot \hat{g}$) or Adam
return θ and baseline values $b(s)$

- Vanilla policy gradient is **on-policy**, i.e., update uses only data from the current policy.

- How to generalize policy gradient to **off-policy**, i.e., update can reuse data from other, past policies?

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Off-policy Version of Policy Gradient

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

- What if we want to use samples collected from a different policy, π ?

$$J(\theta) = E_{\tau \sim p_\theta(\tau)} [r(\tau)]$$

$$J(\theta) = E_{\tau \sim \bar{p}(\tau)} \left[\frac{p_\theta(\tau)}{\bar{p}(\tau)} r(\tau) \right]$$

$$\frac{p_\theta(\tau)}{\bar{p}(\tau)} = \frac{p(s_1) \prod_{t=1}^T \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t)}{p(s_1) \prod_{t=1}^T \bar{\pi}(a_t | s_t) p(s_{t+1} | s_t, a_t)} = \frac{\prod_{t=1}^T \pi_\theta(a_t | s_t)}{\prod_{t=1}^T \bar{\pi}(a_t | s_t)}$$

Say we want to update our latest policy $\pi_{\theta'}$ but we want to use samples from π_θ

$$\nabla_{\theta'} J(\theta') = \mathbb{E}_{\tau \sim p_{\theta'}(\tau)} \left[\left(\sum_{t=1}^T \nabla_{\theta'} \log \pi_{\theta'}(a_t | s_t) \right) \left(\left(\sum_{t'=t}^T r(s_{t'}, a_{t'}) \right) - b \right) \right]$$

$$\nabla_{\theta'} J(\theta') = \mathbb{E}_{\tau \sim p_\theta(\tau)} \left[\frac{p_{\theta'}(\tau)}{p_\theta(\tau)} \left(\sum_{t=1}^T \nabla_{\theta'} \log \pi_{\theta'}(a_t | s_t) \right) \left(\left(\sum_{t'=t}^T r(s_{t'}, a_{t'}) \right) - b \right) \right]$$

$$\nabla_{\theta'} J(\theta') = \mathbb{E}_{\tau \sim p_\theta(\tau)} \left[\prod_{t=1}^T \frac{\pi_{\theta'}(a_t | s_t)}{\pi_\theta(a_t | s_t)} \left(\sum_{t=1}^T \nabla_{\theta'} \log \pi_{\theta'}(a_t | s_t) \right) \left(\left(\sum_{t'=t}^T r(s_{t'}, a_{t'}) \right) - b \right) \right]$$

This can become very small or very large, for larger T

Importance sampling

Using proposal distribution q

$$\begin{aligned} E_{x \sim p(x)} [f(x)] &= \int p(x) f(x) dx \\ &= \int \frac{q(x)}{q(x)} p(x) f(x) dx \\ &= \int q(x) \frac{p(x)}{q(x)} f(x) dx \\ &= E_{x \sim q(x)} \left[\frac{p(x)}{q(x)} f(x) \right] \end{aligned}$$

Note: Important for q to have non-zero support for high probability $p(x)$

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Off-policy Version of Policy Gradient

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

Say we want to update our latest policy $\pi_{\theta'}$ but we want to use samples from π_{θ}

$$\nabla_{\theta'} J(\theta') = \mathbb{E}_{\tau \sim p_{\theta}(\tau)} \left[\left(\prod_{t=1}^T \frac{\pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t)}{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)} \right) \left(\sum_{t=1}^T \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_t | \mathbf{s}_t) \right) \left(\left(\sum_{t'=t}^T r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) - b \right) \right]$$

This can become very small or very large, for larger T

What if we consider the expectation over *timesteps* instead of trajectories?

$$\nabla_{\theta'} J(\theta') \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \frac{\pi_{\theta'}(\mathbf{s}_{i,t}, \mathbf{a}_{i,t})}{\pi_{\theta}(\mathbf{s}_{i,t}, \mathbf{a}_{i,t})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t}) \left(\left(\sum_{t'=t}^T r(\mathbf{s}_{i,t'}, \mathbf{a}_{i,t'}) \right) - b \right)$$

Much less likely to explode/vanish ...but, hard to measure $\frac{\pi_{\theta'}(\mathbf{s}_{i,t})}{\pi_{\theta}(\mathbf{s}_{i,t})}$

Common final form

$$\nabla_{\theta'} J(\theta') \approx \frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T \frac{\pi_{\theta'}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t})}{\pi_{\theta}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t})} \nabla_{\theta'} \log \pi_{\theta'}(\mathbf{a}_{i,t} | \mathbf{s}_{i,t}) \left(\left(\sum_{t'=t}^T r(\mathbf{s}_{i,t'}, \mathbf{a}_{i,t'}) \right) - b \right) \quad \text{often approximated as 1}$$

Full algorithm:

1. sample $\{\tau^i\}$ from $\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$
2. compute $\nabla_{\theta} J(\theta)$ using $\{\tau^i\}$ Can take **multiple gradient steps** on the same batch
3. $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$

One common choice: $\mathbb{E}_{\mathbf{s} \sim \pi_{\theta}} [D_{KL}(\pi_{\theta'}(\cdot | \mathbf{s}) || \pi_{\theta}(\cdot | \mathbf{s}))] \leq \delta$

Run policy to collect batch of data

Improve policy using batch of data

- If a policy **changes too much** before sampling new data, the data no longer reflects states the policy will visit.
- Need to **constrain** the policy not to change too much.

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Reducing Variance with a Critic

<https://web.stanford.edu/class/cs234/slides/lecture5pre.pdf>; https://cs224r.stanford.edu/slides/03_cs224r_policy_gradients_2025.pdf

- Recall the policy gradient update:

$$\nabla_{\theta} \mathbb{E}[R] \approx (1/m) \sum_{i=1}^m \sum_{t=0}^{T-1} \nabla_{\theta} \log \pi_{\theta}(a_t, s_t) (G_t^{(i)} - b(s_t))$$

- The MC rollout return G_t^i is an unbiased estimate of the Q-function, but with a high variance.
- We can instead use a critic to better estimate the Q-function to reduce its variance.
 - **Critic**: Updates Q-function with a parameter w , which is policy evaluation through approximation.
 - **Actor**: Updates policy function using the Q-function produced by critic.
 - **Baseline**: Use the average reward $V(s_t) = \mathbb{E}_{\mathbf{a}_t \sim \pi_{\theta}(\cdot | \mathbf{s}_t)} [Q(s_t, \mathbf{a}_t)]$
- We only need to estimate the difference between Q-function and the baseline, called the **advantage function**, which can be approximated by the TD-error.

$$\begin{aligned} A^{\pi_{\theta}}(s, a) &= Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s) \\ \nabla_{\theta} J(\theta) &= \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) A^{\pi_{\theta}}(s, a)] \end{aligned} \quad \begin{aligned} \delta^{\pi_{\theta}} &= r + \gamma V^{\pi_{\theta}}(s') - V^{\pi_{\theta}}(s) \\ \mathbb{E}_{\pi_{\theta}} [\delta^{\pi_{\theta}} | s, a] &= \mathbb{E}_{\pi_{\theta}} [r + \gamma V^{\pi_{\theta}}(s') | s, a] - V^{\pi_{\theta}}(s) \\ &= Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s) \\ &= A^{\pi_{\theta}}(s, a) \end{aligned}$$

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Compatible Function Approximation Theorem

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-7-policy-gradient-methods.pdf>

Theorem (Compatible Function Approximation Theorem)

If the following two conditions are satisfied:

- 1 Value function approximator is **compatible** to the policy

$$\nabla_w Q_w(s, a) = \nabla_\theta \log \pi_\theta(s, a)$$

- 2 Value function parameters w minimise the mean-squared error

$$\varepsilon = \mathbb{E}_{\pi_\theta} [(Q^{\pi_\theta}(s, a) - Q_w(s, a))^2]$$

Then the policy gradient is exact,

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(s, a) Q_w(s, a)]$$

- Q-function approximation has errors. Does it affect the policy gradient updates?

- This theorem says that not really if the error satisfies some properties.

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Simple Action-Value Actor-Critic Algorithm

<https://davidstarsilver.wordpress.com/wp-content/uploads/2025/04/lecture-7-policy-gradient-methods.pdf>

- Use a linear value function approximation: $Q_w(s, a) = \psi(s, a)^T w$
 - Critic: Update w by a linear TD(0).
 - Actor: Update θ by policy gradient.

Algorithm 2 Simple QAC

- 1: **for** each step **do**
 - 2: generate sample s, a, r, s', a' following π_θ
 - 3: $\delta = r + \gamma Q_w(s', a') - Q_w(s, a)$ #TD error
 - 4: $w \leftarrow w + \beta \delta \psi(s, a)$
 - 5: $\theta \leftarrow \theta + \alpha \nabla_\theta \log \pi_\theta(s, a) Q_w(s, a)$
 - 6: **end for**
-

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A3C Algorithm

<https://lilianweng.github.io/posts/2018-04-08-policy-gradient/>

- Asynchronous advantage actor-critic (A3C) is a policy gradient method with **parallel training (of actors)**.
 - Critics learn the value function while **multiple actors are trained in parallel**.

1. We have global parameters, θ and w ; similar thread-specific parameters, θ' and w' .
2. Initialize the time step $t = 1$.
3. While $T \leq T_{MAX}$:
 1. Reset gradient: $d\theta = 0$ and $dw = 0$.
 2. Synchronize thread-specific parameters with global ones: $\theta' = \theta$ and $w' = w$.
 3. $t_{start} = t$ and sample a starting state s_t .
4. While ($s_t \neq \text{TERMINAL}$) and $t - t_{start} \leq t_{max}$:
 1. Pick the action $A_t \sim \pi_{\theta'}(A_t|S_t)$ and receive a new reward R_t and a new state s_{t+1} .
 2. Update $t = t + 1$ and $T = T + 1$.

5. Initialize the variable that holds the return estimation

$$R = \begin{cases} 0 & \text{if } s_t \text{ is TERMINAL} \\ V_{w'}(s_t) & \text{otherwise} \end{cases}$$

6. For $i = t - 1, \dots, t_{start}$: 1. $R \leftarrow \gamma R + R_i$; here R is a MC measure of G_i . 2. Accumulate gradients w.r.t. θ' : $d\theta \leftarrow d\theta + \nabla_{\theta'} \log \pi_{\theta'}(a_i|s_i)(R - V_{w'}(s_i))$; Accumulate gradients w.r.t. w' : $dw \leftarrow dw + 2(R - V_{w'}(s_i))\nabla_{w'}(R - V_{w'}(s_i))$.
7. Update asynchronously θ using $d\theta$, and w using dw .

<https://github.com/dennybritz/reinforcement-learning/tree/master/PolicyGradient/a3c>

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Asynchronous methods for deep reinforcement learning

V Mnih, AP Badia, M Mirza, A Graves... - ... machine learning, 2016 - proceedings.mlr.press
... DQN was trained on a single Nvidia K40 GPU while the asynchronous methods were ...
asynchronous methods we average over the best 5 models from 50 experiments with learning ...
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DRL for Inventory Control

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Can Deep Reinforcement Learning Improve Inventory Management? Performance on Lost Sales, Dual-Sourcing, and Multi-Echelon Problems

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Abstract. *Problem definition:* Is deep reinforcement learning (DRL) effective at solving inventory problems? *Academic/practical relevance:* Given that DRL has successfully been applied in computer games and robotics, supply chain researchers and companies are interested in its potential in inventory management. We provide a rigorous performance evaluation of DRL in three classic and intractable inventory problems: lost sales, dual sourcing, and multi-echelon inventory management. *Methodology:* We model each inventory problem as a Markov decision process and apply the Asynchronous Advantage Actor-Critic (A3C) DRL algorithm for a variety of parameter settings. *Results:* We demonstrate that the A3C algorithm can match the performance of the state-of-the-art heuristics and other approximate dynamic programming methods. Although the initial tuning was computationally demanding and time demanding, only small changes to the tuning parameters were needed for the other studied problems. *Managerial implications:* Our study provides evidence that DRL can effectively solve stationary inventory problems. This is especially promising when problem-dependent heuristics are lacking. Yet, generating structural policy insight or designing specialized policies that are (ideally provably) near optimal remains desirable.

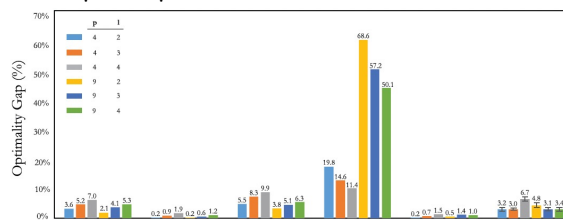
Supplemental Material: The online appendix available at <https://doi.org/10.1287/msom.2021.1064>.

Keywords: inventory theory and control • logistics and transportation • supply chain management • OI-information technology interface

- Some inventory control problems are **notoriously challenging**: lost-sales, dual-sourcing, multi-echelon, etc.

- A3C algorithm matches the performance of **well-designed heuristics**, with limited changes to the tuning parameters.

- Generating **structural policy insight** and specialized near optimal policies remains desirable.



Can deep reinforcement learning improve inventory management? Performance on lost sales, dual-sourcing, and multi-echelon problems

J. Gijbrecchts, R.N. Boute... - ... Management, 2022 - pubsonline.informs.org

... , no topic is hotter today than machine learning and artificial intelligence. We focus on deep reinforcement learning (DRL), the subfield of machine learning that develops "prescriptions" ...

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Agenda

- Value Function Approximation
- DQN
- Policy-based Methods
- Modern DRL
- RL for LLM

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Recap of Value and Policy Based RL

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- **Value-based RL**: First estimate the Q-function, and then generate the deterministic policy using greedy:

$$a_t = \arg \max_a Q(a, s_t)$$

- **Policy-based RL**: First parameterize the stochastic policy output $\pi_\theta(a|s)$, and then optimize the corresponding objective:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R(\tau)]$$

Mostly through **policy gradient**: $\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [G_t \nabla_\theta \log \pi_\theta(s, a)]$

$$\approx \frac{1}{N} \sum_{i=1}^N G_t^i \nabla_\theta \log \pi_\theta(s_i, a_i)$$

- **Actor-Critic**: Estimate the advantage function and parameterize the stochastic policy simultaneously, update both parameters via TD or MC.

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta} [\nabla_\theta \log \pi_\theta(s, a) A^{\pi_\theta}(s, a)]$$

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$$

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Policy Gradient Algorithms

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- 1 The policy gradient has many forms

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) G_t] \text{ — REINFORCE}$$

$$= \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) Q^{\pi}(s, a)] \text{ — Q Actor-Critic}$$

$$= \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) A^{\pi}(s, a)] \text{ — Advantage Actor-Critic}$$

$$= \mathbb{E}_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s, a) \delta] \text{ — TD Actor-Critic}$$

- 2 Critic uses policy evaluation (e.g. MC or TD learning) to estimate $Q^{\pi}(s, a)$, $A^{\pi}(s, a)$, or $V^{\pi}(s)$

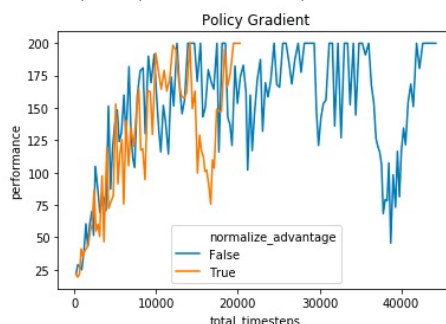
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Issues with Policy Gradient

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- Vanilla policy gradient is **on-policy learning**, so its **sample efficiency is poor**.
- **Large policy update** or **improper step size** will destroy the training.
 - RL is different from supervised learning, where **learning and data are independent**.
 - For RL, if the step is too far $\rightarrow$ bad policy $\rightarrow$ bad data collection, reinforcing each other.
 - May NOT recover from a bad policy, which collapses overall performance.



- Solution: Off-policy learning (importance sampling) + trust region (+ natural policy gradient), **trust region policy optimization (TRPO)**

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Relative Policy Performance

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>: <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- Policy gradient characterizes the steepest ascend in **parameter space** with **Euclidean metric**:

$$d^* = \nabla_{\theta} J(\theta) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \arg \max J(\theta + d), s.t. \|d\| \leq \epsilon$$

- Another perspective is to characterize the steepest ascend in the **distribution space** (of policy output) with **KL-divergence** as the constraint:

$$d^* = \arg \max J(\theta + d), s.t. KL(\pi_{\theta} || \pi_{\theta+d}) = c \quad KL(\pi_{\theta} || \pi_{\theta'}) = E_{\pi_{\theta}}[\log \pi_{\theta}] - E_{\pi_{\theta}}[\log \pi_{\theta'}]$$

$$\begin{aligned} J(\pi') - J(\pi) &= \mathbb{E}_{\tau \sim \pi'} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi}(s_t, a_t) \right] \\ &= \frac{1}{1-\gamma} \mathbb{E}_{\substack{s \sim d^{\pi'} \\ a \sim \pi'}} [A^{\pi}(s, a)] \end{aligned}$$

$$KL(\pi_{\theta} || \pi_{\theta+d}) \approx \frac{1}{2} d^T F d$$

where F is the **Fisher Information Matrix** as second order derivative of KL divergence $E_{\pi_{\theta}}[\nabla \log \pi_{\theta} \nabla \log \pi_{\theta}^T]$

$$d^{\pi}(s) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s | \pi)$$

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Relative Policy Performance

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>: <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

$$\begin{aligned} \mathcal{J}(\theta) - \mathcal{J}(\theta_0) &= \mathcal{J}(\theta) - \mathbb{E}_{s_0 \sim p_0} [V^{\pi_{\theta_0}}(s_0)] \\ &= \mathcal{J}(\theta) - \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} [V^{\pi_{\theta_0}}(s_0)] \\ &= \mathcal{J}(\theta) - \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t V^{\pi_{\theta_0}}(s_t) - \sum_{t=1}^{T-1} \gamma^t V^{\pi_{\theta_0}}(s_t) \right] \\ &= \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right] + \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t (\gamma V^{\pi_{\theta_0}}(s_{t+1}) - V^{\pi_{\theta_0}}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t (r_t + \gamma V^{\pi_{\theta_0}}(s_{t+1}) - V^{\pi_{\theta_0}}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t (Q^{\pi_{\theta_0}}(s_t, a_t) - V^{\pi_{\theta_0}}(s_t)) \right] \\ &= \mathbb{E}_{\tau \sim (p_0, \pi_{\theta}, p)} \left[\sum_{t=0}^{T-1} \gamma^t A^{\pi_{\theta_0}}(s_t, a_t) \right] \end{aligned}$$

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Importance Sampling

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- Apply the standard **importance sampling** technique:

$$\begin{aligned}
 \mathcal{J}(\theta) - \mathcal{J}(\theta_0) &= \mathbb{E}_{\tau \sim (p_0, \pi_\theta, p)} \left[\sum_{t=0}^{T-1} \gamma^t A^{\pi_{\theta_0}}(s_t, a_t) \right] \\
 &= \mathbb{E}_{\substack{s_0 \sim p_0 \\ a_0 \sim \pi_\theta(\cdot | s_0)}} \left[A^{\pi_{\theta_0}}(s_0, a_0) + \mathbb{E}^{\pi_\theta} \left[\sum_{t=1}^{T-1} \gamma^t A^{\pi_{\theta_0}}(s_t, a_t) \mid s_0, a_0 \right] \right] \\
 &= \mathbb{E}_{\substack{s_0 \sim p_0 \\ a_0 \sim \pi_\theta(\cdot | s_0) \\ a'_0 \sim \pi_{\theta_0}(\cdot | s_0)}} \left[\frac{\pi_\theta(a'_0 | s_0)}{\pi_{\theta_0}(a'_0 | s_0)} A^{\pi_{\theta_0}}(s_0, a'_0) + \mathbb{E}^{\pi_\theta} \left[\sum_{t=1}^{T-1} \gamma^t A^{\pi_{\theta_0}}(s_t, a_t) \mid s_0, a_0 \right] \right] \\
 &= \mathbb{E}_{\substack{\tau \sim (p_0, \pi_\theta, p) \\ a'_t \sim \pi_{\theta_0}(\cdot | s_t)}} \left[\sum_{t=0}^{T-1} \gamma^t \frac{\pi_\theta(a'_t | s_t)}{\pi_{\theta_0}(a'_t | s_t)} A^{\pi_{\theta_0}}(s_t, a'_t) \right]
 \end{aligned}$$

State sampled from the **new policy**, action sampled from the **original policy**.

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Surrogate Objective

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- If the new policy is **sufficiently close** to the original one:

$$\mathcal{J}(\theta) = \mathbb{E}_{\substack{\tau \sim (p_0, \pi_\theta, p) \\ a'_t \sim \pi_{\theta_0}(\cdot | s_t)}} \left[\sum_{t=0}^{T-1} \gamma^t \frac{\pi_\theta(a'_t | s_t)}{\pi_{\theta_0}(a'_t | s_t)} A^{\pi_{\theta_0}}(s_t, a'_t) \right] + C \quad \mathcal{K}(\theta; \theta_0) = \mathbb{E}_{\substack{\tau \sim (p_0, \pi_{\theta_0}, p) \\ a'_t \sim \pi_{\theta_0}(\cdot | s_t)}} \left[\sum_{t=0}^{T-1} \gamma^t \frac{\pi_\theta(a'_t | s_t)}{\pi_{\theta_0}(a'_t | s_t)} A^{\pi_{\theta_0}}(s_t, a'_t) \right] + C$$

State and action sampled from the **original policy**.

- For $\theta \approx \theta_0$,

$$\mathcal{K}(\theta; \theta_0) = \mathbb{E}_{\tau \sim (p_0, \pi_{\theta_0}, p)} \left[\sum_{t=0}^{T-1} \gamma^t \frac{\pi_\theta(s_t, a_t)}{\pi_{\theta_0}(s_t, a_t)} A^{\pi_{\theta_0}}(s_t, a_t) \right]$$

Also denoted as
 $\mathcal{L}_{\theta_0}(\theta)$

$$\stackrel{(*)}{\approx} \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T^{(i)}-1} \gamma^t \frac{\pi_\theta(s_t^{(i)}, a_t^{(i)})}{\pi_{\theta_0}(s_t^{(i)}, a_t^{(i)})} \hat{A}_t^{(i)} \quad \text{Estimate of advantage function}$$

- We could maximize the surrogate objective subject to a trust-region constraint:

$$\begin{aligned}
 &\underset{\theta \in \mathbb{R}^p}{\text{maximize}} && \frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T^{(i)}-1} \gamma^t \frac{\pi_\theta(s_t^{(i)}, a_t^{(i)})}{\pi_{\theta_0}(s_t^{(i)}, a_t^{(i)})} \hat{A}_t \\
 &\text{subject to} && \theta \text{ and } \theta_0 \text{ close}
 \end{aligned}$$

Needed for
 $\mathcal{K}(\theta, \theta_0) \approx \mathcal{J}(\theta)$ and $(*)$

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TRPO

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>
<https://spinningup.openai.com/en/latest/algorithms/trpo.html>; <https://lilianweng.github.io/posts/2018-04-08-policy-gradient/>

Algorithm 3 Trust Region Policy Optimization

Input: initial policy parameters θ_0

for $k = 0, 1, 2, \dots$ **do**

Collect set of trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$

Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm

Form sample estimates for

- policy gradient $\hat{g}_k$ (using advantage estimates)
- and KL-divergence Hessian-vector product function $f(v) = \hat{H}_k v$

Use CG with n_{cg} iterations to obtain $x_k \approx \hat{H}_k^{-1} \hat{g}_k$

Estimate proposed step $\Delta_k \approx \sqrt{\frac{2\delta}{x_k^T \hat{H}_k x_k}} x_k$

Perform backtracking line search with exponential decay to obtain final update

$$\theta_{k+1} = \theta_k + \alpha^j \Delta_k$$

Operationalize the trust-region constraint
 $KL(\pi_{\theta_{old}}(\cdot|s_t) || \pi_{\theta}(\cdot|s_t)) \leq \delta$

end for

Trust region policy optimization

J. Schulman, S. Levine, P. Abbeel, ... on machine learning, 2015 - proceedings.mlr.press

... a practical algorithm, called Trust Region Policy Optimization (TRPO). This algorithm is effective ... Despite its approximations that deviate from the theory, TRPO tends to give monotonic ...

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Code: https://github.com/openai/baselines/tree/master/baselines/trpo_mpi

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Limitations of TRPO

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- TRPO has scalability issues:
 - Computing the **Fisher Information Matrix** H for the current policy is expensive, because it requires a large batch of rollouts to approximate H .

$$H = \nabla_{\theta}^2 KL(\pi_{\theta_t} || \pi_{\theta}) = E_{a,s \sim \pi_{\theta_t}} \left[\nabla_{\theta} \log \pi_{\theta}(a, s) \nabla_{\theta} \log \pi_{\theta}(a, s)^T \right]$$

- Determine the step-size of gradient descent involves performing an **approximate Newton method**, which is similar to natural gradient method, with $H^{-1}g$ solved via a **conjugate gradient (CG)** solver. For details, see the original TRPO paper.

Then the objective turns to:

$$J_{\theta_t}(\theta) \approx g^T(\theta - \theta_t)$$

$$\theta_{t+1} = \arg \max_{\theta} g^T(\theta - \theta_t) \text{ s.t. } \frac{1}{2}(\theta - \theta_t)^T H(\theta - \theta_t) \leq \delta$$

$KL(\theta_t || \theta) \approx \frac{1}{2}(\theta - \theta_t)^T H(\theta - \theta_t)$ This is a quadratic equation and can be solved analytically:

$$\theta_{t+1} = \theta_t + \sqrt{\frac{2\delta}{g^T H^{-1} g}} H^{-1} g$$

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Proximal Policy Optimization (PPO)

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- The trust region is modeled as the constraint in TRPO:

$$\begin{aligned} & \text{maximize}_{\theta} \mathbb{E}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} A_t \right] \\ & \text{subject to } \mathbb{E}_t [KL[\pi_{\theta_{old}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]] \leq \delta \end{aligned}$$

- We can also rewrite it as an **unconstrained optimization**, so **KL is a regularizer**:

$$\text{maximize}_{\theta} \mathbb{E}_t \left[\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} A_t \right] - \beta \mathbb{E}_t [KL[\pi_{\theta_{old}}(\cdot | s_t), \pi_{\theta}(\cdot | s_t)]]$$

- This is called PPO-penalty, we only need to compute the 1st-order derivative of KL-divergence.
- The penalty coefficient is **adaptive** (β_k) between iterations to **approximately enforce KL-divergence constraint**.

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PPO-Penalty

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

Algorithm 4 PPO with Adaptive KL Penalty

Input: initial policy parameters θ_0 , initial KL penalty β_0 , target KL-divergence δ

for $k = 0, 1, 2, \dots$ **do**

Collect set of partial trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$

Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm

Compute policy update

$$\theta_{k+1} = \arg \max_{\theta} \mathcal{L}_{\theta_k}(\theta) - \beta_k \bar{D}_{KL}(\theta || \theta_k)$$

by taking K steps of minibatch SGD (via Adam)

if $\bar{D}_{KL}(\theta_{k+1} || \theta_k) \geq 1.5\delta$ **then**

$\beta_{k+1} = 2\beta_k$

else if $\bar{D}_{KL}(\theta_{k+1} || \theta_k) \leq \delta/1.5$ **then**

$\beta_{k+1} = \beta_k/2$

end if

end for

PPO has a **similar performance** as TRPO but can be solved **much faster** by SGD/Adam.

Proximal policy optimization algorithms

J. Schulman, F. Wolski, P. Dhariwal, A. Radford... - arXiv preprint arXiv ..., 2017 - arxiv.org

... It shows how several objectives vary as we interpolate along the **policy** update direction, obtained by **proximal policy optimization** (the **algorithm** we will introduce shortly) on a ...

☆ 保存 弉 引用 被引用次数: 36481 相关文章 所有 12 个版本 弉

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PPO-Clip

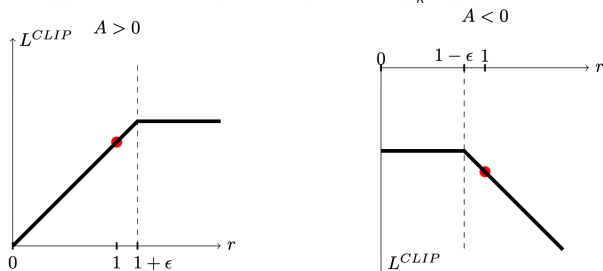
<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>

- Define the probability ratio: $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_k}(a_t|s_t)$
 - $r_t(\theta)$ is **close to 1** $\rightarrow$ the new policy is close to the original one.
- PPO-Clip utilizes a clipped surrogate objective to ensure that the policy does not change much:

$$\mathcal{L}_{\theta_k}^{CLIP}(\theta) = \mathbb{E}_{\tau \sim \pi_k} \left[\sum_{t=0}^T \left[\min(r_t(\theta) \hat{A}_t^{\pi_k}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^{\pi_k}) \right] \right]$$

where ϵ is a hyperparameter (maybe $\epsilon = 0.2$)

Policy update is $\theta_{k+1} = \arg \max_{\theta} \mathcal{L}_{\theta_k}^{CLIP}(\theta)$



The final objective is a **lower/pessimistic bound** of the unclipped objective.

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PPO-Clip

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>
<https://spinningup.openai.com/en/latest/algorithms/ppo.html>; <https://lilianweng.github.io/posts/2018-04-08-policy-gradient/>

Algorithm PPO with Clipped Objective

Input: initial policy parameters θ_0 , clipping threshold ϵ

for $k = 0, 1, 2, \dots$ **do**

Collect set of partial trajectories $\mathcal{D}_k$ on policy $\pi_k = \pi(\theta_k)$

Estimate advantages $\hat{A}_t^{\pi_k}$ using any advantage estimation algorithm

Compute policy update

$$\theta_{k+1} = \arg \max_{\theta} \mathcal{L}_{\theta_k}^{CLIP}(\theta)$$

by taking K steps of minibatch SGD (via Adam), where

$$\mathcal{L}_{\theta_k}^{CLIP}(\theta) = \mathbb{E}_{\tau \sim \pi_k} \left[\sum_{t=0}^T \left[\min(r_t(\theta) \hat{A}_t^{\pi_k}, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t^{\pi_k}) \right] \right]$$

end for

- Clipping prevents the policy from having incentive to go far away from θ_k .
- PPOs have the stability and reliability of TRPO but are **much simpler to implement**.

Code: <https://github.com/openai/baselines/tree/master/baselines/ppo1>

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PPO Implementation is Tricky and Challenging

<https://web.stanford.edu/class/cs234/slides/lecture6pre.pdf>; <https://ernestryu.com/courses/RL-LLM/chapter1.pdf>
<https://spinningup.openai.com/en/latest/algorithms/ppo.html>; <https://lilianweng.github.io/posts/2018-04-08-policy-gradient/>

Revisiting Design Choices in Proximal Policy Optimization

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Abstract

Proximal Policy Optimization (PPO) is a popular deep policy gradient algorithm. In standard implementations, PPO regularizes policy updates with clipped probability ratios, and parameterizes policies with either continuous Gaussian distributions or discrete Softmax distributions. These design choices are widely accepted, and motivated by empirical performance comparisons on MuJoCo and Atari benchmarks.

We revisit these practices outside the regime of current benchmarks, and expose three failure modes of standard PPO. We explain why standard design choices are problematic in these cases, and show that alternative choices of surrogate objectives and policy parameterizations can prevent the failure modes. We hope that our work serves as a reminder that many algorithmic design choices in reinforcement learning are tied to specific simulation environments. We should not implicitly accept these choices as a standard part of a more general algorithm.

The failure modes are:

1. On continuous action spaces, standard PPO is unstable when rewards vanish outside bounded support.
2. On discrete action spaces with sparse high rewards, standard PPO often gets stuck at suboptimal actions.
3. The policy is sensitive to initialization when there are locally optimal actions close to initialization.

Code: <https://github.com/openai/baselines/tree/master/baselines/ppo1>

IMPLEMENTATION MATTERS IN DEEP POLICY GRADIENTS: A CASE STUDY ON PPO AND TRPO

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ABSTRACT

We study the roots of algorithmic progress in deep policy gradient algorithms through a case study on two popular algorithms: Proximal Policy Optimization (PPO) and Trust Region Policy Optimization (TRPO). Specifically, we investigate the consequences of “code-level optimizations:” algorithm augmentations found only in implementations or described as auxiliary details to the core algorithm. Seemingly of secondary importance, such optimizations turn out to have a major impact on agent behavior. Our results show that they (a) are responsible for most of PPO’s gain in cumulative reward over TRPO, and (b) fundamentally change how RL methods function. These insights show the difficulty and importance of attributing performance gains in deep reinforcement learning.

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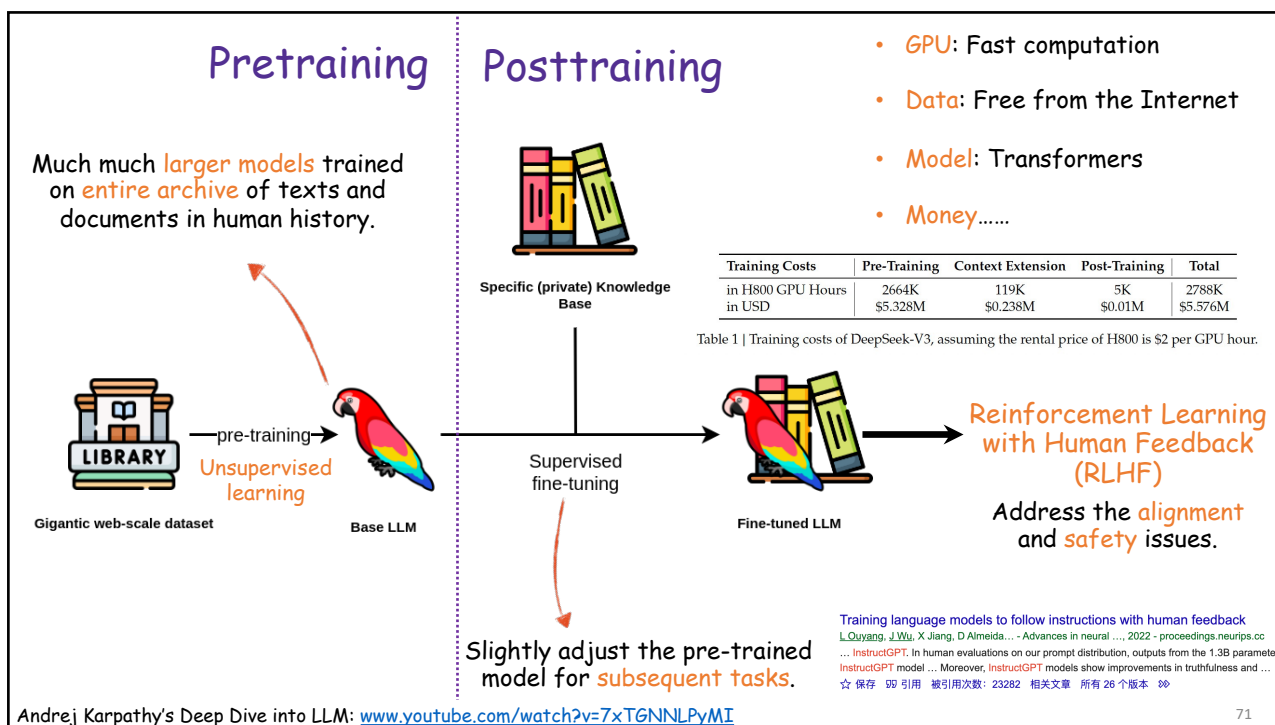
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Agenda

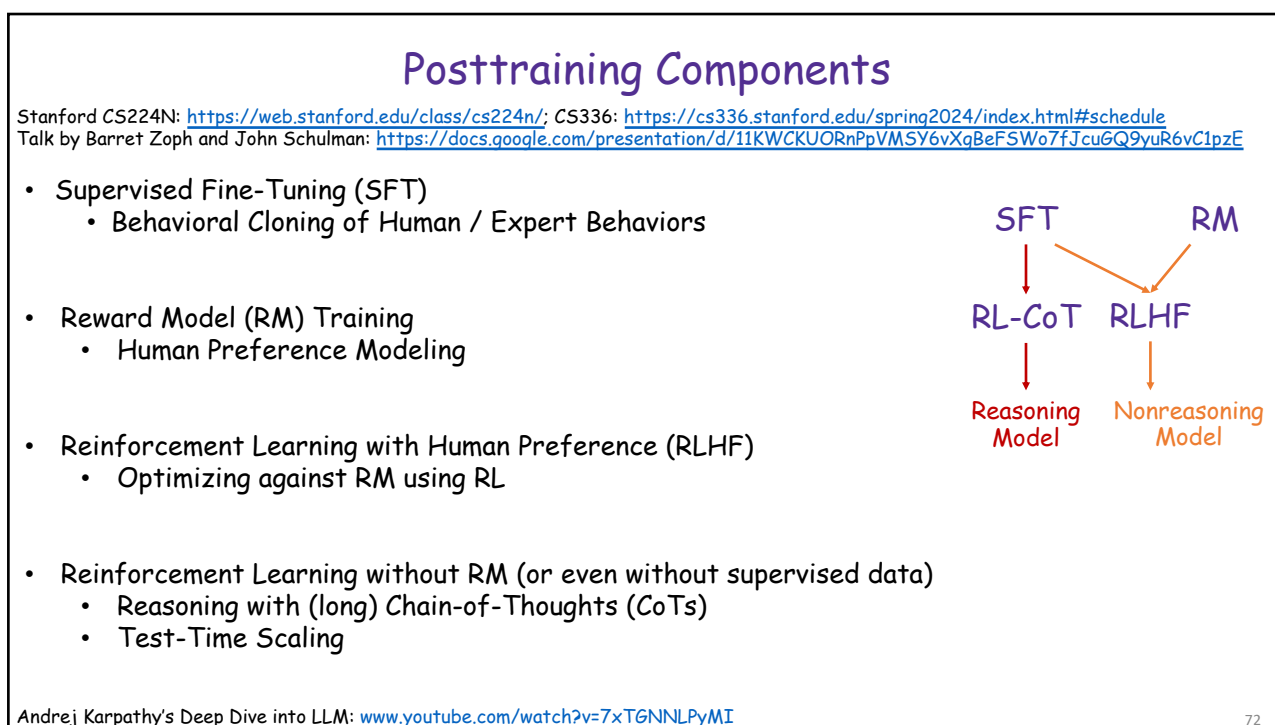
- Value Function Approximation
- DQN
- Policy-based Methods
- Modern DRL
- RL for LLM

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Reward Modeling

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
Talk by Barret Zoph and John Schulman: <https://docs.google.com/presentation/d/11KWCKUORnPPVMSY6vXqBeFSWo7fJcu6Q9yuR6vC1pzE>

• Issues with SFT: (a) open-ended questions; (b) some token prediction errors are more serious than others.

• Human labellers are asked to rank K LLM-generated responses to a prompt.

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}} E_{(x, y_w, y_l) \sim D} [\log(\sigma(r_\theta(x, y_w) - r_\theta(x, y_l)))]$$

Prompt Win Response Lose Response Sigmoid Reward Model (RM), usually a "Small Language Model" (GPT-3 6B)

• RM helps generalize LLM evaluations for difficult to verify tasks.

• Saves huge costs to recruit human labellers.

• RMs are subject to reward-hacking.

Rank analysis of incomplete block designs: I. the method of paired comparisons
RA Bradley, ME Terry - Biometrika, 1952 - JSTOR
... possibility of extending the analysis to incomplete block designs with larger block sizes. We
... The method of paired comparisons becomes inefficient where it is possible to rank more ...
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Lilian Weng's Blog: <https://lilianweng.github.io/posts/2024-11-28-reward-hacking/>

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Reinforcement Learning with Human Feedback (RLHF)

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
Talk by Barret Zoph and John Schulman: <https://docs.google.com/presentation/d/11KWCKUORnPPVMSY6vXqBeFSWo7fJcu6Q9yuR6vC1pzE>

Write a story about frogs

Proximal Policy Optimization

Once upon a time...

RM

r_k

Prompts Dataset

Initial Language Model

Tuned Language Model (RL Policy)

Reinforcement Learning Update (e.g. PPO)

$\theta \leftarrow \theta + \nabla_{\theta} J(\theta)$

Reward (Preference) Model

$r_{\theta}(y|x)$

Critic Network

Challenge: RLHF is quite unstable.

Once with an RM, we use RL to automatically optimize the output of fine-tuned LLM in alignment with human preferences.

objective (ϕ) = $E_{(x,y) \sim D_{\pi_{\phi}^{\text{RL}}}} [r_{\theta}(x,y) - \beta \log(\pi_{\phi}^{\text{RL}}(y|x)/\pi^{\text{SFT}}(y|x))] + \gamma E_{x \sim D_{\text{pretrain}}} [\log(\pi_{\phi}^{\text{RL}}(x))] + \text{Loss on the pretraining distribution}$

Not too far away from SFT

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RL of RLHF

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- RLHF finetunes the LLM π further so that the completion achieves high reward as measured by the reward model $r(\cdot)$.
- What is the MDP in RLHF?
 - LLMs are autoregressive models, so they are not Markovian but history dependent: the policy is not of the form $\pi(a_t|s_t)$ but something like $\pi(a_t|s_1, s_2, \dots, s_t)$.
 - Therefore, the state should be the **user prompt and the tokens generated** so far $(u_1, u_2, \dots, u_l)$.
 - Each action is the generation of one token. The policy is randomized, but the transition dynamics is **deterministic**.

The RL setup

- Each timestep is a BPE token.
- The LLM $\pi_\theta(u_{\ell+1}|u_1, \dots, u_\ell)$ is our policy mapping current state $(u_1, \dots, u_\ell)$ to a distribution on the action (next token) $u_{\ell+1}$.
- Response generation is an episode, and an episode terminates when LM generates $\langle \text{EOS} \rangle$.
- No discount used, i.e., discount factor $\gamma = 1$ is used.
- Reward (by reward model r_ψ) is only provided at the end of the episode. There are no intermediate reward. Called "contextual bandit" setting.
- Sampling temperature $\beta = 1$.

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PPO for RLHF

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Let $\pi_\theta(u_{\ell+1}|u_1, \dots, u_\ell)$ be our LLM and the RL policy.
- Let x be a text prompt and $y = y_{1:T+1}$ be its completion by π_θ . (So $y_{T+1} = \langle \text{EOS} \rangle$.)
- Let $y_{1:t}$ the partial completion up to token t .
- The PPO-clip ratio is set to $\epsilon = 0.2$.

PPO maintains a value function model $V_\phi(x, y_{1:t})$: Given $(x, y_{1:t})$, what is the expected reward if we continue generation with π_θ .

Advantage $\hat{A} = r_\psi(x, y_{1:T+1}) - V_\phi(x, y_{1:t})$: How good is the total completion $y_{t+1:T+1}$ compared to what V_ϕ was expecting based on $y_{1:t}$?

If $\hat{A} = r_\psi(x, y_{1:T+1}) - V_\phi(x, y_{1:t}) > 0$, then $y_{t+1:T+1}$ was a good completion. We should adjust π_θ to make those actions more likely.

If $\hat{A} = r_\psi(x, y_{1:T+1}) - V_\phi(x, y_{1:t}) < 0$, then $y_{t+1:T+1}$ was a bad completion. We should adjust π_θ to make those actions less likely.

(Actually GAE was used for $\hat{A}$, but let's consider the simpler advantage estimate for simplicity.)

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PPO v0 for RLHF (subjective to reward hacking)

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

while (not converged)

sample N trajectories $(x^{(i)}, y_{1:T^{(i)}+1}^{(i)}) \sim (p_0, \pi_{\theta_{\text{curr}}}^{\text{RL}})$ for $i = 1, \dots, N$

$\hat{A}_t^{(i)} = r_\psi(x^{(i)}, y_{1:T^{(i)}+1}^{(i)}) - V_\phi(x^{(i)}, y_{1:t}^{(i)})$ for $i = 1, \dots, N, t = 0, \dots, T^{(i)}$

solve:

$$\underset{\theta_{\text{next}} \in \mathbb{R}^p}{\text{maximize}} \quad \sum_{i=1}^N \sum_{t=0}^{T^{(i)}} \mathcal{C}_\epsilon \left(\frac{\pi_{\theta_{\text{next}}}^{\text{RL}}(y_{t+1}^{(i)} | x^{(i)}, y_{1:t}^{(i)})}{\pi_{\theta_{\text{curr}}}^{\text{RL}}(y_{t+1}^{(i)} | x^{(i)}, y_{1:t}^{(i)})}, \hat{A}_t^{(i)} \right)$$

$\theta_{\text{curr}} = \theta_{\text{next}}$

solve:

$$\underset{\phi \in \mathbb{R}^q}{\text{minimize}} \quad \frac{1}{N} \sum_{i=1}^N \frac{1}{T^{(i)}} \sum_{t=0}^{T^{(i)}} \frac{1}{2} (r_\psi(x^{(i)}, y_{1:T^{(i)}+1}^{(i)}) - V_\phi(x^{(i)}, y_{1:t}^{(i)}))^2$$

end

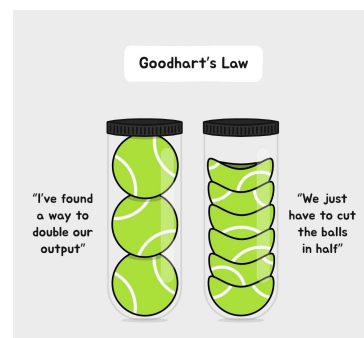
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Reward Hacking

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Goodhart's Law: When a measure becomes a target, it ceases to become a good measure.
- PPO v0 suffers from over-optimization/reward hacking
 - Reward model is **imperfect**, maximizing reward model too much will exploit the model imperfections and result in **adversarial generations**.
 - The reward model is trained on the SFT LLM π_{SFT} . Therefore, the reward model is informative about responses generated by the RL policy only if it π_{RL} is **close** to π_{SFT} .
- Therefore, moving away from π_{SFT} too much will cause the language model to lose its main capabilities.
 - **Fine-tuning too much** can break the model, causing it to output nonsense tokens.
- **Solution**: Impose a KL-divergence penalty term, ensuring that π_{RL} is **close** to π_{SFT} .



Lilian Weng's Blog on Reward Hacking: <https://lilianweng.github.io/posts/2024-11-28-reward-hacking/>

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KL-penalty and Pre-training Loss

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- RLHF with KL-penalty maximizes the objective:

$$\begin{aligned}
 \mathcal{J}(\theta) &= \mathbb{E}_{(x,y) \sim D_{\pi_{\theta}^{\text{RL}}}} \left[r_{\psi}(x, y) \right] - \underbrace{\beta \mathbb{E}_{x \sim \mathcal{D}} \left[D_{\text{KL}}(\pi_{\theta}^{\text{RL}}(\cdot | x) \| \pi^{\text{SFT}}(\cdot | x)) \right]}_{\text{KL-penalty}} \\
 &= \mathbb{E}_{(x,y) \sim D_{\pi_{\theta}^{\text{RL}}}} \left[r_{\psi}(x, y) - \beta \log \frac{\pi_{\theta}^{\text{RL}}(y | x)}{\pi^{\text{SFT}}(y | x)} \right] \\
 &= \mathbb{E}_{(x,y) \sim D_{\pi_{\theta}^{\text{RL}}}} \left[r_{\psi}(x, y) - \beta \sum_{t=0}^T \log \frac{\pi_{\theta}^{\text{RL}}(y_{t+1} | x, y_{1:t})}{\pi^{\text{SFT}}(y_{t+1} | x, y_{1:t})} \right]
 \end{aligned}$$

- The KL-penalty encourages π_{RL} to stay **close** to π_{SFT} .

Lilian Weng's Blog on Reward Hacking: <https://lilianweng.github.io/posts/2024-11-28-reward-hacking/>

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KL-penalty and Pre-training Loss

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- The new objective function:

$$\mathcal{J}(\theta) = \mathbb{E}_{(x,y) \sim D_{\pi_{\theta}^{\text{RL}}}} \left[r_{\psi}(x, y) - \beta \sum_{t=0}^T \log \frac{\pi_{\theta}^{\text{RL}}(y_{t+1} | x, y_{1:t})}{\pi^{\text{SFT}}(y_{t+1} | x, y_{1:t})} \right]$$

- Maximizing $\mathcal{J}(\theta)$ is equivalent to RL on MDP with the same transition dynamics with modified rewards $r_0, r_1, \dots, r_T$ given by

$$\begin{aligned}
 r_t &= -\beta \log \frac{\pi_{\theta}^{\text{RL}}(y_{t+1} | x, y_{1:t})}{\pi^{\text{SFT}}(y_{t+1} | x, y_{1:t})}, \quad t = 0, \dots, T-1 \\
 r_T &= r_{\psi}(x, y_{1:T+1}) - \beta \log \frac{\pi_{\theta}^{\text{RL}}(y_{T+1} | x, y_{1:T})}{\pi^{\text{SFT}}(y_{T+1} | x, y_{1:T})}
 \end{aligned}$$

- The modified MDP has **immediate rewards**, which **absorb the KL penalty**.
- The reward depends on the parameter θ , but all the math goes through.

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KL-penalty and Pre-training Loss

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Even with the KL-penalty, the **base language model capability is compromised**, so we add another **LLM pretraining loss**:

$$\mathcal{J}(\theta) = \mathbb{E}_{(x,y) \sim D_{\text{RL}}} \left[r_{\psi}(x,y) - \beta \log \left(\pi_{\theta}^{\text{RL}}(y|x) / \pi^{\text{SFT}}(y|x) \right) \right] + \eta \mathbb{E}_{x \sim D_{\text{pretrain}}} \left[\log \left(\pi_{\theta}^{\text{RL}}(x) \right) \right]$$

- The last term is the next-token prediction loss used in pre-training, where $\eta > 0$.

while (not converged)

sample N trajectories $(x^{(i)}, y_{1:T(i)+1}^{(i)}) \sim (p_0, \pi_{\theta_{\text{curr}}}^{\text{RL}})$ for $i = 1, \dots, N$

$$\hat{A}_t^{(i)} = r_{\psi}(x^{(i)}, y_{1:T(i)+1}^{(i)}) - \beta \sum_{s=t}^{T(i)} \log \left(\frac{\pi_{\theta_{\text{next}}}^{\text{RL}}(y_{s+1}^{(i)} | x^{(i)}, y_{1:s}^{(i)})}{\pi_{\theta_{\text{curr}}}^{\text{RL}}(y_{s+1}^{(i)} | x^{(i)}, y_{1:s}^{(i)})} \right) - V_{\phi}(x^{(i)}, y_{1:t}^{(i)}) \quad \text{for } i=1, \dots, N, t=0, \dots, T(i)$$

$= \hat{G}_t^{(i)}$

solve:

$$\underset{\theta_{\text{next}} \in \mathbb{R}^p}{\text{maximize}} \quad \sum_{i=1}^N \sum_{t=0}^{T(i)} \mathcal{C}_{\varepsilon} \left(\frac{\pi_{\theta_{\text{next}}}^{\text{RL}}(y_{t+1}^{(i)} | x^{(i)}, y_{1:t}^{(i)})}{\pi_{\theta_{\text{curr}}}^{\text{RL}}(y_{t+1}^{(i)} | x^{(i)}, y_{1:t}^{(i)})}, \hat{A}_t^{(i)} \right)$$

$\theta_{\text{curr}} = \theta_{\text{next}}$

solve:

$$\underset{\phi \in \mathbb{R}^q}{\text{minimize}} \quad \frac{1}{N} \sum_{i=1}^N \frac{1}{T(i)} \sum_{t=0}^{T(i)} \frac{1}{2} (\hat{G}_t^{(i)} - V_{\phi}(x^{(i)}, y_{1:t}^{(i)}))^2$$

$$\mathcal{C}_{\varepsilon}(\ell, A) = \min(\ell A, \text{clip}_{1-\varepsilon}^{1+\varepsilon}(\ell) A)$$

PPO Algorithm with KL-penalty

PPO and pre-training updates are performed **simultaneously** or in **alternating** fashion.

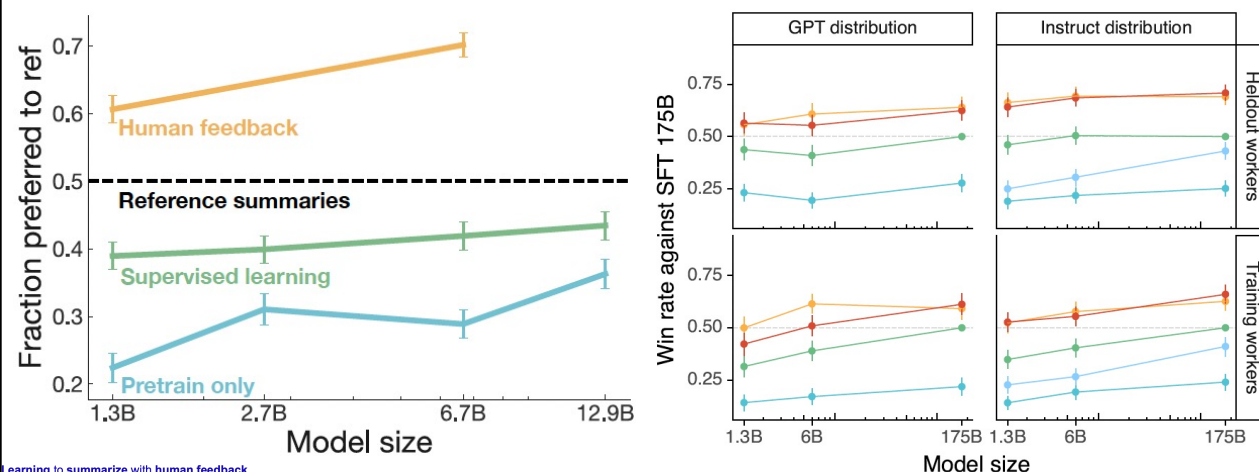
end

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Empirical Performance of RLHF

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Learning to summarize with human feedback

N. Stiennon, L. Ouyang, J. Wu, et al. - Advances in neural information processing systems, 2020

... original human demonstrations in the dataset (see Figure 1). (2) We show human feedback

... Our RL-trained human feedback models also generate high-quality summaries of news ...

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Training language models to follow instructions with human feedback

L. Ouyang, J. Wu, X. Jiang, D. Almeida, et al. - Advances in neural information processing systems, 2022

... InstructGPT. In human evaluations on our prompt distribution, outputs from the 1.3B parameter

InstructGPT model ... Moreover, InstructGPT models show improvements in truthfulness and ...

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Learning to Summarize: <https://arxiv.org/pdf/2009.01325>; InstructGPT: <https://arxiv.org/pdf/2203.02155>

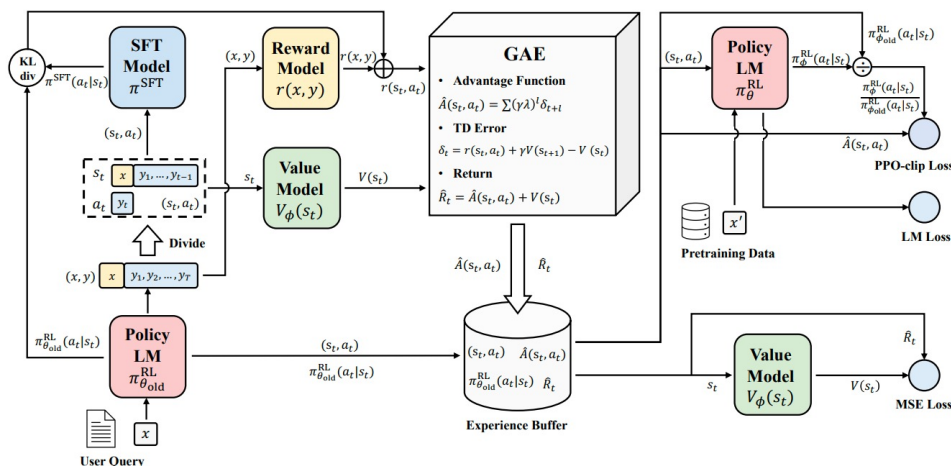
82

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RLHF Can Be (Too) Complex

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- RL optimization is **computationally expensive and tricky**.
- You need to **fit value functions**.
- **Online sampling** is slow.
- Performance is highly **sensitive to hyperparameters**.



Secrets of RLHF (Part I): <https://arxiv.org/abs/2307.04964>

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Direct Preference Optimization (DPO)

Stanford CS224N: <https://web.stanford.edu/class/cs224n/>; CS336: <https://cs336.stanford.edu/spring2024/index.html#schedule>
 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Standard RLHF:
 1. Fit the reward model based on human preference data.
 2. Optimize the instruction-finetuned LLM given the reward function.
- Another perspective:
 1. Derive the reward model based on an RL policy, parameterized by θ .
 2. Optimize the parameter θ by **fitting the reward model to the true human preference data**, instead of a given reward model.
- Interestingly, there is a **closed-form solution** for the optimal KL-regularized policy given any reward model. Ignoring the pre-training loss, the KL-regularized RL solves:

$$\underset{\theta}{\text{maximize}} \quad \mathcal{J}(\pi_{\theta}; r) = \mathbb{E}_{\substack{x \sim \mathcal{D} \\ y \sim \pi_{\theta}}} \left[r(x, y) - \beta \log \frac{\pi_{\theta}(y | x)}{\pi^{\text{SFT}}(y | x)} \right]$$

r is any **reward model**.

Direct preference optimization: Your language model is secretly a reward model

R Rafailov, A Sharma, E Mitchell, ... - Advances in neural ... - 2023 - proceedings.neurips.cc
 ... of human **preferences** and then use RL to **optimize** a language ... **optimize** a language model to adhere to human **preferences**. ... We propose **Direct Preference Optimization (DPO)**, an ...

Direct Preference Optimization: <https://arxiv.org/pdf/2305.18290>

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Closed-form Solution for KL-regularized RL

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Transforming the objective as a loss:

$$\begin{aligned} -\frac{1}{\beta} \mathcal{J}(\pi_\theta; r) &= \mathbb{E}_{\substack{x \sim \mathcal{D} \\ y \sim \pi_\theta}} \left[\log \frac{\pi_\theta(y|x)}{\pi^{\text{SFT}}(y|x)} - \frac{1}{\beta} r(x, y) \right] \\ &= \mathbb{E}_{\substack{x \sim \mathcal{D} \\ y \sim \pi_\theta}} \left[\log \frac{\pi_\theta(y|x)}{\pi^{\text{SFT}}(y|x)} - \log \exp \left(\frac{1}{\beta} r(x, y) \right) + \log Z(x) - \log Z(x) \right] \\ &= \mathbb{E}_{\substack{x \sim \mathcal{D} \\ y \sim \pi_\theta}} \left[\log \frac{\pi_\theta(y|x)}{\pi_r(y|x)} - \log Z(x) \right] = \mathbb{E}_{x \sim \mathcal{D}} \left[D_{\text{KL}}(\pi_\theta(\cdot|x) \parallel \pi_r(\cdot|x)) - \log Z(x) \right] \end{aligned}$$

where $\pi_r(y|x) = \frac{\pi^{\text{SFT}}(y|x) \exp(\frac{1}{\beta} r(x, y))}{Z(x)}$, $Z(x) = \sum_y \pi^{\text{SFT}}(y|x) \exp(\frac{1}{\beta} r(x, y))$

- Therefore, maximize $\mathcal{J}(\pi_\theta; r)$ is equivalent to minimize $\mathbb{E}_{x \sim \mathcal{D}} [D_{\text{KL}}(\pi_\theta(\cdot|x) \parallel \pi_r(\cdot|x))]$, i.e., the optimal policy is $\pi_\theta = \pi_r$.

- Inversely, the reward function that makes any policy π optimal is:

$$r_\pi(x, y) = \beta \log \frac{\pi(y|x)}{\pi^{\text{SFT}}(y|x)} + \beta \log Z(x).$$

Direct preference optimization: Your language model is secretly a reward model

R Rafailov, A Sharma, E Mitchell... - Advances in neural ... 2023 - proceedings.neurips.cc
 ... of human preferences and then use RL to optimize a language ... optimize a language model to adhere to human preferences. ... We propose **Direct Preference Optimization (DPO)**, an ...

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Direct Preference Optimization:

<https://arxiv.org/pdf/2305.18290>

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DPO

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

- Recall that the reward model was trained via the BTL method:

$$\underset{\psi}{\text{minimize}} \sum_{\substack{(x, y_i, y_j) \in \mathcal{D} \\ y_i > y_j}} -\log \sigma(r_\psi(x, y_i) - r_\psi(x, y_j))$$

- DPO substitutes:

$$r_\psi \leftarrow r_\pi(x, y) = \beta \log \frac{\pi(y|x)}{\pi^{\text{SFT}}(y|x)} + \beta \log Z(x).$$

- Therefore, DPO trains the neural net that solves:

$$\underset{\theta}{\text{minimize}} \sum_{\substack{(x, y_i, y_j) \in \mathcal{D} \\ y_i > y_j}} -\log \sigma \left(\beta \log \frac{\pi_\theta(y_i|x)}{\pi^{\text{SFT}}(y_i|x)} - \beta \log \frac{\pi_\theta(y_j|x)}{\pi^{\text{SFT}}(y_j|x)} \right)$$

Direct preference optimization: Your language model is secretly a reward model

R Rafailov, A Sharma, E Mitchell... - Advances in neural ... 2023 - proceedings.neurips.cc
 ... of human preferences and then use RL to optimize a language ... optimize a language model to adhere to human preferences. ... We propose **Direct Preference Optimization (DPO)**, an ...

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DPO

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

$$\underset{\theta}{\text{minimize}} \quad \underbrace{\sum_{\substack{(x, y_i, y_j) \in \mathcal{D} \\ y_i > y_j}} -\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_i | x)}{\pi^{\text{SFT}}(y_i | x)} - \beta \log \frac{\pi_{\theta}(y_j | x)}{\pi^{\text{SFT}}(y_j | x)} \right)}_{=\mathcal{L}^{\text{DPO}}(\theta)}$$

- There's no need to train a reward model + there's no need to train the value network.
- DPO effectively converts the RL problem into a **supervised learning problem**, so it is **much easier to execute**.
- There is debate on DPO vs. PPO.

Gradient: $\nabla_{\theta} \mathcal{L}^{\text{DPO}}(\theta) = -\beta \sum_{\substack{(x, y_i, y_j) \in \mathcal{D} \\ y_i > y_j}} \underbrace{\sigma(-r_{\pi_{\theta}}(x, y_i) - r_{\pi_{\theta}}(x, y_j))}_{=\text{reward model error}} \left(\underbrace{\nabla_{\theta} \log \pi_{\theta}(y_i | x)}_{\text{increase likelihood of } y_i} - \underbrace{\nabla_{\theta} \log \pi_{\theta}(y_j | x)}_{\text{decrease likelihood of } y_j} \right)$

- **Interpretation:** DPO is doing ascent on the good outcome y_i , while doing descent on the bad completion y_j . The gradient is accentuated if the implicitly defined reward model $r_{\pi_{\theta}}$ disagrees with the human preference $y_i > y_j$.

Is dpo superior to ppo for llm alignment? a comprehensive study

S.Xu, W.Fu, J.Gao, W.Ye, W.Liu, Z.Mei, G.Wang... - arXiv preprint arXiv ..., 2024 - arxiv.org

... In this paper, we focus more on the performance of DPO and PPO. We observe that DPO-iter performs better than DPO but worse than PPO. PPO consistently achieves a higher reward ...

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DPO vs. PPO: <https://arxiv.org/abs/2404.10719>

87

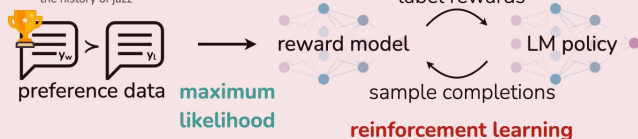
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Empirical Performance of DPO

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 UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

Reinforcement Learning from Human Feedback (RLHF)

x: "write me a poem about the history of jazz"

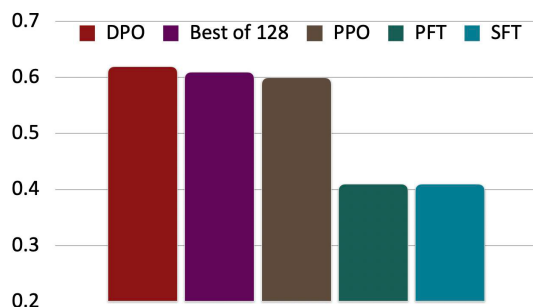


Direct Preference Optimization (DPO)

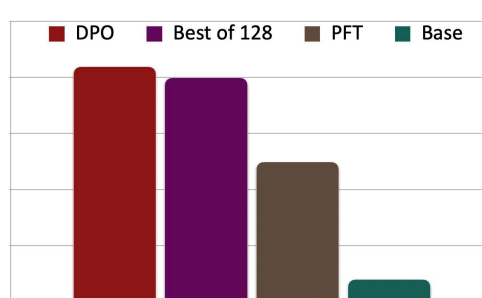
x: "write me a poem about the history of jazz"



Summarization helpfulness win rates vs ground truth



Dialogue helpfulness win rates vs ground truth



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DPO in Action: Balancing Engagement and Polarization

- The **engagement** and **polarization** scores are annotated by **GPT-4o mini**.

Balancing Engagement and Polarization: Multi-Objective Alignment of News Content Using LLMs

Mengjie (Magie) Cheng⁺⁺ Elie Ofek⁺ Hema Yoganarasimhan[#]
⁺Harvard Business School
[#]University of Washington

Abstract

We study how media firms can use LLMs to generate news content that aligns with multiple objectives – making content more engaging while maintaining a preferred level of polarization/slant consistent with the firm's editorial policy. Using news articles from *The New York Times*, we first show that more engaging human-written content tends to be more polarizing. Further, naively employing LLMs (with prompts or standard Direct Preference Optimization approaches) to generate more engaging content can also increase polarization. This has an important managerial and policy implication: using LLMs without building in controls for limiting slant can exacerbate news media polarization. We present a constructive solution to this problem based on the Multi-Objective Direct Preference Optimization (MODPO) algorithm, a novel approach that integrates Direct Preference Optimization with multi-objective optimization techniques. We build on open-source LLMs and develop a new language model that simultaneously makes content more engaging while maintaining a preferred editorial stance. Our model achieves this by modifying content characteristics strongly associated with polarization but that have a relatively smaller impact on engagement. Our approach and findings apply to other settings where firms seek to use LLMs for content creation to achieve multiple objectives, e.g., advertising and social media.

Keywords: LLMs, Content Creation, Digital Marketing, Media, News, Polarization, Generative AI, DPO

Table 2: Relationship between Polarization and Engagement in the Original Data

	(1) Engagement	(2) Polarization
Constant	1.092*** (0.0227)	1.070*** (0.0177)
Topics	No	Yes
N	13,508	13,508
R ²	0.1427	0.5725

Standard errors in parentheses
⁺ p < 0.05, ⁺⁺ p < 0.01, ⁺⁺⁺ p < 0.001

$$\mathcal{L}_{\text{MODPO}}(\pi_{\theta_w}; \mathbf{r}_{\phi, -k}, \pi_{\text{slf}}, \mathcal{D}_k) = -\mathbb{E}_{\mathcal{D}_k} \left[\log \sigma \left(\frac{\beta}{\mathbf{w}_k} \log \frac{\pi_{\theta_w}(\mathbf{y}_w | \mathbf{x})}{\pi_{\text{slf}}(\mathbf{y}_w | \mathbf{x})} - \frac{\beta}{\mathbf{w}_k} \log \frac{\pi_{\theta_w}(\mathbf{y}_l | \mathbf{x})}{\pi_{\text{slf}}(\mathbf{y}_l | \mathbf{x})} \right) - \frac{1}{\mathbf{w}_k} \mathbf{w}_k^T (\mathbf{r}_{\phi, -k}(\mathbf{x}, \mathbf{y}_w) - \mathbf{r}_{\phi, -k}(\mathbf{x}, \mathbf{y}_l)) \right]$$

margin, $m_{\phi}(\mathbf{x}, \mathbf{y}_w, \mathbf{y}_l)$

Table 3: Summary Statistics of Engagement and Polarization for More and Less Engaging Versions

	More Engaging Version	Less Engaging Version	More Engaging Version	Less Engaging Version
Mean	9.00	3.00	7.00	3.00
Std	8.99	3.10	5.89	3.09
N	13,508	13,508	13,508	13,508

* p < 0.05, ** p < 0.01, *** p < 0.001

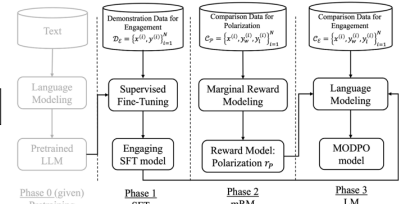


Figure 2: Overview of MODPO Implementation

Table 5: Gemini Evaluation Results (with Engaging SFT as the Reference Policy)

	Difference in Engagement Scores			Difference in Polarization Scores		
	$\Delta_{\text{MODPO.ORG}}$	$\Delta_{\text{EDPO.ORG}}$	$\Delta_{\text{MODPO.EDPO}}$	$\Delta_{\text{MODPO.ORG}}$	$\Delta_{\text{EDPO.ORG}}$	$\Delta_{\text{MODPO.EDPO}}$
Mean	2.13***	2.57***	-0.44***	0.80***	1.75***	-0.95***
Std	1.15	1.58	1.28	1.66	2.08	1.37
Median	2.00	3.00	-1.00	1.00	2.00	-1.00
N	983.00	983.00	983.00	983.00	983.00	983.00

* p < 0.05, ** p < 0.01, *** p < 0.001.

$\Delta_{\text{MODPO.ORG}}$ = MODPO version - Original version; $\Delta_{\text{EDPO.ORG}}$ = Engaging DPO version - Original version; $\Delta_{\text{MODPO.EDPO}}$ = MODPO version - Engaging DPO version.

Beyond one-preference-fits-all alignment: Multi-objective direct preference optimization
 Z. Zhou, L. Liu, J. Shen, X. Yan, C. Yang, - Findings of the ... 2024 - aclanthology.org
 A single language model, even when aligned with labelers through reinforcement learning from human feedback (RLHF), may not suit all human preferences. Recent approaches therefore prefer customization, gathering multi-dimensional feedback, and creating distinct reward models for each dimension. Different language models are then optimized for various preferences using multi-objective RLHF (MODPO) with varying reward weights. However, RL fine-tuning is unstable and resource-heavy, especially with diverse and ...
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Balancing Engagement and Polarization: <https://arxiv.org/pdf/2504.13444>; Multi-objective DPO: <https://aclanthology.org/2024.findings-acl.630/>

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Chain-of-thought prompting elicits reasoning in large language models
 J. Wei, X. Wang, Q. Schuermans, - Advances in neural ... 2022 - proceedings.neurips.cc
 ... a chain of thought—a series of intermediate reasoning steps—of large language models to perform complex reasoning. In ... such reasoning abilities emerge naturally in sufficiently large ...
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UCLA RL for LLM: <https://ernestryu.com/courses/RL-LLM/chapter3.pdf>

Chain-of-Thought (CoT)

Large language models are zero-shot reasoners
 T. Kojima, S.S. Gu, M. Reid, Y. Matsuo, - Advances in neural ... 2022 - proceedings.neurips.cc
 ... show that LLMs are decent zero-shot reasoners by simply adding "Let's ... zero-shot-CoT mainly to standard CoT without prompting to verify the effectiveness of its chain of thought reasoning.
 ☆ 保存 ☆ 引用 被引用次数: 8307 相关文章 所有 10 个版本 30

- Chain-of-thought (CoT) is a technique for LLMs to **think aloud to itself** before producing an answer.

- CoT **greatly improves the performance** compared with immediately producing an answer.

- An easy way to induce CoT: "Let's think step by step".

- CoT can also be induced by **in-context prompting**.

- Modern LLMs are **instruction-finetuned** to exhibit the CoT behavior.

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
 A: The answer is 11.
 Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
 A:
 (Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
 A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.
 Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
 A:
 (Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. The answer is 4. ✓

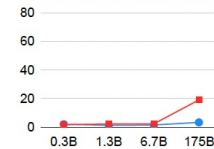
(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
 A: The answer (arabic numerals) is
 (Output) 8 ✗

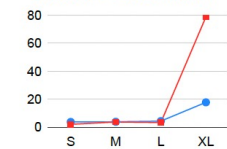
(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
 A: Let's think step by step.
 (Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

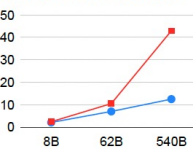
(a) MultiArith on Original GPT-3



(b) MultiArith on Instruct GPT-3



(c) GMS8K on PaLM



Zero-shot reasoners: <https://arxiv.org/pdf/2205.11916>; CoT elicits reasoning: <https://arxiv.org/pdf/2201.11903>

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